When Are Two Heads Better than One? The Organizational Design of Innovation*

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Abstract

We study the optimal allocation of tasks over the life cycle of an innovative project. Innovation requires two complementary activities, research and development, both subject to moral hazard. Agents also privately observe the quality of discoveries, so the principal must elicit truthful reports to avoid costly mistakes. Assigning both tasks to a single "solo operator" avoids free-riding but creates a double-deviation problem: the agent can shirk on research and conceal the resulting low-quality discovery. As time passes without a discovery, this deviation becomes increasingly attractive. Separating the tasks across two agents eliminates the double deviation but introduces free-riding. We characterize how this tradeoff shapes organization over time. When low-quality discoveries are readily exposed, the project is run by sequential solo operators; when they are difficult to expose, a two-agent team is optimal from the outset. At intermediate levels of detectability, the project begins with a solo operator and switches endogenously to a team.

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"But of these two thynges he wolde determyne none
Without aide. For two hedds are better than one."

—John Heywood, 1546

1 Introduction

Innovation projects combine two broad activities: searching for promising ideas and transforming them into working products or services. However, organizations differ not only in how they divide these activities, but also in when they do so. Some projects begin as a "one-person shop," with a founder or lone engineer exploring the technological frontier while also building the first working version of the product, and divide responsibilities only as the project evolves (Levy, 2011). Others separate exploratory research from execution-oriented development from the outset, assigning the activities to units with distinct objectives and evaluation criteria (O'Reilly and Tushman, 2004). Still others expand conditionally as progress unfolds, bringing in short-term consultants to diagnose setbacks (Brown, 2008) or activating specialized "tiger teams" when an initial approach disappoints (Pavlak, 2004). These arrangements reflect different choices about whether to concentrate responsibility in a generalist, rely on specialists, or add expertise only after the project reaches a critical stage.

These patterns pose an organizational-design puzzle. Technological comparative advantage can explain why different tasks are assigned to different specialists, but it cannot by itself explain why an organization reallocates unchanged tasks across equally capable agents as a project unfolds. Incentive considerations provide a distinct explanation. Effort in innovation is difficult to observe, and the people producing a discovery often learn its quality before the principal decides whether to accept it; a flaw may become verifiable only later through the realized payoff. Task allocation therefore determines who controls effort, who possesses information, and which deviations are feasible. Bundling both tasks with one agent can economize on incentive payments, but it allows that agent to combine shirking with misreporting. Separating the tasks limits this joint deviation but creates scope for free-riding. We therefore ask how a principal should allocate research and development, when a second agent should be hired, and whether changing incentive conditions alone can generate a transition from a solo operator to a specialized team.

To isolate the organizational tradeoff, we begin with a static benchmark stripped of any technological reason for specialization. A project requires two complementary tasks: *research*, which increases the likelihood that a discovery is high quality, and *development*,

which produces the discovery itself. We call a high-quality discovery a 'peach" and a low-quality discovery a 'lemon." A peach surely yields a positive payoff when accepted. A lemon yields the positive payoff with probability $1-q$, but with probability q acceptance generates a large loss, exposing low quality. Thus, q measures the ex post verifiability of quality: when q is high, lemons are easy to expose; when q is low, they are difficult to distinguish from peaches. The principal can bundle both tasks with a single *solo operator* or separate them between a researcher and a developer. The agents are identical and equally productive at either task, so the two organizational forms are technologically equivalent. Any difference between them arises entirely from incentives.

With a solo operator, the same agent controls research effort, development effort, and the report of quality. This concentration of responsibility avoids free-riding across agents, but it creates a particularly profitable joint deviation. The agent can shirk on research, continue working on development, and, if the resulting discovery is a lemon, report it as a peach. We call this combination of shirking and misreporting the *double deviation*. When q is high, such deception is likely to be exposed, and the relevant constraint is instead *double shirking* – shirking on both tasks. In that region, the principal need only compensate the agent for the private benefits he forgoes by working; i.e., his opportunity cost. As q falls, however, endeavoring to pass off a lemon as a peach becomes increasingly attractive. Deterring the double deviation then requires additional rents, so the incentive cost of a solo operator rises as quality becomes harder to verify.

Separating the tasks removes this information rent altogether. Suppose the researcher controls the effort that affects quality while the developer generates the discovery and reports its quality. Because the developer's effort affects whether a discovery is produced but not whether it is a peach or a lemon, the principal can give him the same compensation following either report. Truthful reporting therefore requires no additional rent. At the same time, the researcher no longer controls the quality report and hence cannot conceal shirking by misreporting a lemon. In fact, the result is stronger: the minimum cost of employing the two-agent team is exactly the same as it would be if quality were publicly observable. *Separating the tasks makes private information effectively costless to the principal.*

This informational advantage comes at a price however. Research improves the probability that a discovery is a peach but is not necessary for a discovery to be produced. A shirking researcher can therefore free-ride on the developer: development still produces a discovery, and even without research it may turn out to be a peach. The researcher consequently retains some prospect of being rewarded even when he shirks, so inducing research effort requires compensation above his private benefit from shirking. Unlike the solo operator's information rent, this free-riding cost does not depend on q. The static organizational choice

therefore reduces to a transparent tradeoff. When lemons are easy to expose, bundling is cheaper because the solo operator avoids free-riding and misreporting is inexpensive to deter. When lemons are difficult to expose, separation is cheaper because the team eliminates the information rent associated with the double deviation. Thus, whether two heads are better than one depends on whether the informational benefit of separating tasks outweighs the free-riding cost it creates.

The static benchmark isolates the two forces that govern organizational design: bundling research and development avoids free-riding but creates an information rent, while separating the tasks eliminates that information rent at the cost of free-riding. Our main model adds the dynamic feature that makes this organizational choice evolve over the life of an innovative project. Discoveries arrive according to a Poisson process as long as development effort continues. Continuous research effort ensures that an arriving discovery is a peach; without research, the discovery may instead be a lemon. The key new element is that the incentive cost of a solo operator now depends on the history of the project. Early on, the prospect of future employment gives the solo operator valuable continuation utility. Because passing off a lemon as a peach risks exposure, termination, and the loss of this future utility, continuation value acts as a *bond* that can deter the double deviation at no additional cost. But this bond is gradually depleted as the project continues without a discovery. As the solo operator's continuation value falls, misreporting becomes harder to deter and eventually requires additional rents. By contrast, a two-agent team continues to eliminate the information problem, while its free-riding cost remains stationary. Dynamics therefore transform the static comparison into an endogenous organizational life cycle: a solo operator can be optimal early in a project, when continuation value provides effective discipline, yet become more expensive than a team after a sufficiently long spell without success.

This history dependence generates three distinct organizational regimes. When lemons are easy to detect ex post, the project is best organized around a succession of solo innovators: one agent performs both research and development until he is replaced by another. When lemons are difficult to detect, the informational cost of bundling the tasks is sufficiently severe that the principal instead begins immediately with a two-agent team, assigning research and development to different agents. The most interesting case, the staggered overlap regime, arises at intermediate levels of detectability. The project begins with a solo innovator, whose continuation value initially provides an effective bond against misreporting. As that bond erodes, however, the principal brings in a second agent before replacing the first, and the organization transitions endogenously from a solo operation to a specialized team. Thus, the model predicts not only whether innovation is organized around one head or two, but also when a second head should enter as the project evolves.

Within the staggered-overlap regime, the transition from one head to two can take two distinct forms. In the *External Consultant* structure, the incumbent remains responsible for development while the entrant immediately specializes in research. In the *Standby Developer* structure, by contrast, the entrant initially takes over both tasks and the incumbent is temporarily placed on standby; later, the incumbent returns to development while the entrant specializes in research. Which transition is optimal depends on the value of preserving the incumbent's incentives. Near the boundary at which simultaneous hiring becomes optimal, the organization adopts the External Consultant structure. Farther from that boundary, when maintaining the incumbent's stake in the project is sufficiently valuable, the Standby Developer structure emerges instead. Thus, even conditional on eventually moving to a two-agent team, the model determines not only *when* the second agent should enter, but also *how* responsibilities should be reallocated during the transition.

We make three contributions to the theory of organizational design. First, we identify a new source of history dependence in organizational structure. Even with stationary technology, unchanged tasks, and ex ante identical agents, the optimal organization evolves over the life of a project. The key state variable is the incumbent's continuation utility: the future stake that can be used to discipline current behavior. As unsuccessful search erodes this stake, a solo operator becomes progressively more expensive to incentivize, eventually making a change in organizational form optimal. Performance history therefore affects organization not through learning about project quality, worker ability, or technology, but through the endogenous depletion of incentive capacity. This mechanism generates transitions from a generalist organization to a specialized team even though the underlying production environment never changes.

Second, we uncover a fundamental incentive tradeoff between bundling and separating tasks when both actions and information are private. Bundling research and development with a single agent avoids free-riding across workers, but it also allows that agent to control both the quality of a discovery and its report. This creates a joint deviation in which he shirks on research and hopes to pass off a low-quality discovery as a high-quality one. Separating the tasks eliminates this deviation: because the reporting agent does not control quality, truthful reporting requires no information rent. Indeed, under a two-agent team, private information about discovery quality becomes effectively public from the principal's perspective. The cost of separation is instead free-riding in research, since a researcher who shirks may still benefit from discoveries produced by the developer. Organizational design therefore trades off the information rent created by bundling against the free-riding cost created by separation.

Finally, the model yields a set of practical predictions about when organizations should

rely on generalists, specialists, and transitional arrangements between the two. When low-quality discoveries are likely to be exposed ex post, the threat of losing future continuation value provides effective discipline, allowing the organization to rely longer on solo innovators. When flawed output is difficult to detect, this disciplinary force is weaker, specialization occurs earlier, and a two-agent team may be optimal from the outset. The model also predicts how an additional worker should be deployed during the transition. When preserving the incumbent's remaining incentive stake is especially valuable, the entrant temporarily takes over both tasks while the incumbent is held on standby; closer to the region in which simultaneous hiring is optimal, the entrant instead joins immediately as a research specialist. These predictions distinguish our mechanism from explanations based on comparative advantage, learning-by-doing, or changes in the underlying production technology.

Related Literature. This paper contributes to the literature on team production and task allocation in agency settings. The most closely related paper is Laux (2001), which studies task bundling under limited liability when tasks are independent and their outcomes are separately observable. In that setting, assigning all tasks to a single agent is always optimal, because bundling relaxes the limited-liability constraint: outcomes from tasks on which the agent exerts effort can effectively "bond" the agent and deter shirking on other tasks. Schmitz (2005) studies a related two-stage model that favors separation through a different mechanism. As in our model, success in the first stage is not required for success in the second. When a single agent controls both stages, the most profitable deviation is therefore to shirk in the first stage to increase rent extraction in the second. Separating control rights mitigates this distortion by limiting the second-stage agent's ability to benefit from early failure. However, this mechanism relies on strong assumptions: the outcome of the first stage must be observable and contractible, and control rights must be allocated before first-stage outcomes are realized. If control rights could instead be made contingent on first-stage performance, it would be optimal to allow an agent to continue only following success, analogous to the sequential hiring structure we characterize in Section 5 and to the optimal organization in Laux (2001) when the contracting problem can be repeated after task failure.

In a similar vein, Gromb and Martimort (2007) study a setting in which a principal relies on experts to gather and report two private signals about a project's value. They show that hiring two experts, each responsible for gathering and reporting one signal, can be cheaper than relying on a single agent to report both signals, because contracts can exploit cross-checking: the agent is paid only if the two reports agree. This ranking is reversed, however, when experts can collude, either horizontally among themselves or vertically with

the principal. Related tradeoffs also appear in the literature on public-private partnerships (Martimort and Pouyet, 2008; Iossa and Martimort, 2012, 2015), which studies analogous issues in procurement settings. A key distinction is that these models are static in their contracting structures and control rights. In contrast, we study a dynamic environment in which the optimal organization – one agent versus two – varies endogenously with agents' performance histories, generating a richer set of organizational arrangements consistent with those observed in practice.

Our study is also related to the broader literature on team effort, including Admati and Perry (1991), Keller, Rady, and Cripps (2005), Bonatti and Horner (2011), Cetemen, Hwang, and Kaya (2020), Yildirim (2023), among others. This literature has traditionally focused on moral hazard – specifically, the underprovision of effort by team members – either across agents (e.g., free-riding) or over time (e.g., procrastination). A subset of studies emphasizes the benefits of teams. For example, Bolton and Harris (1999), Yildirim (2006), and Georgiadis (2015) analyze environments in which effort by one team member increases the productivity of others, often by accelerating the arrival of a critical event (e.g., project completion) at which all team members are rewarded. The latter two studies also examine how the project horizon interacts with team size, showing that encouragement effects dominate early in a project, while free-riding becomes more severe at later stages. Other work, e.g., Marx and Matthews (2000) and Che and Yoo (2001), highlights how teams may deter free-riding in dynamic settings when agents can collectively punish deviations by withholding future effort.

Our study differs from this literature in several important ways. First, all of the aforementioned studies consider teams of fixed size.[1] They typically consider identical agents performing identical tasks. In contrast, we allow the principal to adjust the team size dynamically as the project unfolds and identify when the expansion of the team (from one to two agents) is optimal. Moreover, because we consider asymmetric tasks, our results generate additional implications for *how* the additional team member will be utilized, not only *when*. Finally, in most of the existing studies, there is typically no agency friction when the project is operated by a single agent. Instead, agency considerations such as free-riding or procrastination arise from team production itself. In our model, agency frictions exist even with a single agent in the form of both moral hazard and adverse selection. However, the nature of the main agency friction (whether it is under-provision of effort or misreporting of the outcome) determines whether team effort exacerbates or mitigates these frictions.

[1] Yildirim (2006) and Georgiadis (2015) discuss optimal team sizes, but team size must be chosen before project inception and cannot be changed over time. Thus, the costs and benefits of larger teams are derived through comparative statics.

Finally, our model belongs to the dynamic contracting literature with both moral hazard and adverse selection, such as Gershkov and Perry (2012), Halac, Kartik, and Liu (2016), Green and Taylor (2016), He, Wei, Yu, and Gao (2017), Liu and Lu (2018), Varas (2018), Che, Iossa, and Rey (2021), etc. In all these studies, the principal only contracts with a single agent. Our model allows contracting with multiple agents, demonstrating how different organizational arrangements (a solo innovator or an agent team) help reduce the cost of incentives for different constellations of agency frictions. Our model illustrates how moral hazard or adverse selection can endogenously become costly to incentivize relative to each other, both statically and dynamically, and how these shifting incentive costs generate optimal transitions across organizational forms.

Outline. Section 2 presents a static benchmark that highlights the static tradeoff between employing a single agent and an agent team. Section 3 introduces the dynamic environment. Section 4 derives the incentive compatibility conditions under each organizational structure. Section 5 presents our main results on optimal organizational design. Section 6 discusses several model variations. Section 7 concludes. All proofs are in the Appendix.

2 Static Benchmark

2.1 Basic Environment

There are two dates, $t \in \{0, 1\}$, a principal, and two ex ante identical agents. The principal undertakes a project requiring two complementary tasks, which we call *research* and *development*. Either agent is equally capable of performing either task, or both. Thus, there is no technological gain from specialization; any advantage of assigning the tasks to different agents arises entirely from incentives. All players are risk neutral, and agents are protected by limited liability. We refer to the principal as "she" and to each agent as "he."

At $t = 0$, the principal can invest in either or both tasks, each at cost c. Development requires the assigned agent to choose unobservable effort $e \in \{s, w\}$. If he works, the project produces a discovery at $t = 1$; if he shirks, he obtains private benefit $b < c$ and no discovery is generated.

Research affects the quality rather than the arrival of the discovery. The agent assigned to research chooses unobservable effort $a \in \{s, w\}$. If he shirks, he obtains private benefit b, and a discovery generated by development is a high-quality "peach" with probability $\lambda_0 \in (0, 1)$. If he works, he forgoes this private benefit and raises the probability of a peach

to $\lambda_1 \in (\lambda_0, 1)$. Let

$$\theta \equiv \lambda_1 - \lambda_0 > 0 \tag{1}$$

denote the improvement in discovery quality generated by research.

The principal can assign both tasks to a single agent, whom we call a *solo operator*, or assign research and development to different agents, whom we call the *researcher* and *developer*.

At $t = 1$, if a discovery is generated, the agent or agents observe its quality and report it to the principal. The principal then decides whether to accept it. An accepted peach yields $R > 0$. An accepted low-quality "lemon" yields R with probability $1 - q$ but generates a loss $-L < 0$ with probability q, where $q \in (0, 1]$. Thus, q measures the ex post detectability of a lemon: a larger q makes it more difficult to pass off a lemon as a successful discovery.

We impose the following assumption.

Assumption 1 *Lemon discoveries should not be accepted:* $(1 - q)R < qL$. *Research is socially optimal:* $\theta R > c$.

To isolate the organizational tradeoff, we restrict attention to incentive-compatible contracts under which the principal invests in both tasks, the assigned agents exert effort, and quality is reported truthfully.[2] On the equilibrium path, a discovery is therefore generated with probability one and accepted if and only if it is a peach. When indifferent, an agent exerts effort and reports truthfully.

Before solving for optimal contracts, it is useful to preview the organizational tradeoff. Bundling the tasks avoids a free-riding problem because one agent bears responsibility for both research and development. But it also gives that agent control over both the quality of the discovery and the report of its quality. He can therefore combine two deviations: shirk on research and, if a lemon results, conceal his shirking by reporting the lemon as a peach. Separating the tasks eliminates this *double deviation*, because the agent whose effort determines quality need not control the quality report. The price of separation is free-riding: a researcher who shirks can still benefit from a discovery produced by the developer. The optimal organizational form is determined by which of these two incentive costs is larger.

[2]Development is essential, so research is the only optional task. Appendix B.6 provides a sufficient condition under which inducing research effort is optimal; the results in this section apply whenever that condition holds.

2.2 Analysis

A. Solo Operator

Suppose first that a single agent performs both tasks. A contract is a pair (β, γ), where the solo operator receives β if he reports a peach and the accepted discovery yields R, and receives γ if he reports a lemon.

Following the prescribed actions, his expected compensation is

$$U(\beta, \gamma) \equiv \lambda_1 \beta + (1 - \lambda_1)\gamma. \tag{2}$$

There are two economically relevant deviations. First, the agent can shirk on both tasks, obtaining private benefit $2b$ and producing no discovery. Deterring this double shirking requires

$$U(\beta, \gamma) \geq 2b. \tag{Double-Shirking IC}$$

Second, the agent can shirk on research, obtain private benefit b, continue to work on development, and then report the resulting discovery strategically. Deterring this deviation requires

$$U(\beta, \gamma) \geq b + \lambda_0 \beta + (1 - \lambda_0) \max\{\gamma, (1 - q)\beta\}. \tag{3}$$

Truthful reporting of a lemon separately requires

$$\gamma \geq (1 - q)\beta. \tag{Truthtelling IC}$$

At the optimum, the truth-telling constraint binds. Substituting $\gamma = (1 - q)\beta$ into (3) yields

$$U(\beta, \gamma) \geq b + \lambda_0 \beta + (1 - \lambda_0)(1 - q)\beta. \tag{Double-Deviation IC}$$

The deviation on the right-hand side combines shirking on research with misreporting: the agent works on development but reports every discovery as a peach. We therefore call it the *double deviation*. Bundling the two tasks is valuable because it avoids free-riding, but costly because it makes precisely this joint deviation feasible.

The principal's problem is consequently to minimize compensation while deterring both double shirking and the double deviation.

Lemma 1 (Cost of a solo operator) *Let*

$$D(q) \equiv \lambda_1 + (1 - \lambda_1)(1 - q). \tag{4}$$

For a solo operator, the minimum expected compensation for effort on both tasks and truthful reporting is

$$C^{solo}(q) = \max\left\{2b, \frac{bD(q)}{q\theta}\right\}. \tag{5}$$

This cost is implementable by

$$\beta^* = \max\left\{\frac{2b}{D(q)}, \frac{b}{q\theta}\right\}, \quad \gamma^* = (1 - q)\beta^*. \tag{6}$$

The two terms in $C^{solo}(q)$ correspond directly to the two deviations above. They are equal at

$$\bar{q} \equiv \frac{1}{1 + \lambda_1 - 2\lambda_0}. \tag{7}$$

Whenever $\bar{q} \leq 1$,

$$C^{solo}(q) = \begin{cases} \dfrac{bD(q)}{q\theta}, & q < \bar{q}, \\ 2b, & q \geq \bar{q}. \end{cases} \tag{8}$$

When q is high, an accepted lemon is likely to be exposed. Misreporting is therefore unattractive, and double shirking is the relevant deviation. The principal can then induce both tasks at expected cost $2b$, exactly equal to the agent's total private benefit from shirking. In this region, bundling achieves the lowest possible incentive cost.

When q is low, by contrast, a lemon can more easily masquerade as a peach. Shirking on research while continuing development and concealing a bad discovery becomes the more attractive deviation. The principal must then leave the solo operator an information rent to induce both research effort and truthful reporting. Accordingly, $C^{solo}(q)$ is strictly decreasing in q on this branch: making quality harder to verify raises the cost of bundling the tasks.

B. Two-Agent Team

Now suppose that research and development are assigned to different agents. Both agents observe quality, and the principal may solicit the quality report from either or both of them.

The minimum cost is the same regardless of how the reporting responsibility is assigned.

Lemma 2 (Cost of a two-agent team) *The minimum expected compensation required to induce effort on both tasks and truthful reporting from a two-agent team is*

$$C^{team} = \left(\frac{2\lambda_1 - \lambda_0}{\lambda_1 - \lambda_0}\right) b > 2b. \tag{9}$$

No alternative allocation of reporting responsibility or report-contingent compensation can lower this cost.

To see the source of this cost, suppose the developer reports quality. Because his effort determines whether a discovery is produced but has no effect on its quality, the cheapest way to induce both development effort and truthful reporting is

$$\beta_D = \gamma_D = b. \tag{10}$$

The developer is compensated for his foregone shirking benefit but is given no reason to prefer reporting a peach to reporting a lemon.

For the researcher, it is optimal to set $\gamma_R = 0$. His incentive constraint is then

$$\lambda_1 \beta_R \geq b + \lambda_0 \beta_R. \tag{11}$$

At the optimum it binds, giving

$$\beta_R = \frac{b}{\theta} > b. \tag{12}$$

Expected total compensation is therefore

$$C^{team} = \lambda_1(\beta_R + b) + (1 - \lambda_1)b = \left(\frac{2\lambda_1 - \lambda_0}{\lambda_1 - \lambda_0}\right) b. \tag{13}$$

Separation thus has two sharply contrasting effects. Its benefit is informational. The developer controls the report but not the probability that the discovery is a lemon, so the principal can make him indifferent across reports. The researcher controls quality but not the report. Neither agent can combine shirking on research with concealing the resulting lemon. Indeed, C^{team} is exactly the cost the principal would incur if quality were publicly observable: private information creates no additional incentive cost under separation.

Its cost is free-riding. Research is valuable because it raises the probability that a discovery produced by the developer is a peach, but research is not itself necessary for a discovery

to occur. A shirking researcher therefore still benefits from the developer's effort and receives $\lambda_0\beta_R$ with positive probability. Inducing research must compensate him not only for the private benefit b from shirking, but also for this foregone chance of being rewarded while free-riding. Consequently, $C^{team} > 2b$.

Crucially, this free-riding cost is independent of q. Separation removes the connection between the researcher's hidden action and the agent responsible for reporting quality, so making lemons harder to detect does not make the team more expensive to incentivize.

2.3 Cost Comparison

Because both organizational forms implement the same equilibrium actions, the principal's investment costs and expected project revenue are identical under bundling and separation. Organizational design therefore reduces to a comparison of $C^{solo}(q)$ and C^{team}.

Proposition 1 *Define*

$$q^* = \frac{1}{1+\theta}. \tag{14}$$

The principal strictly prefers a two-agent team if $q < q^*$ and strictly prefers a solo operator if $q > q^*$.

Figure 1 illustrates the comparison.

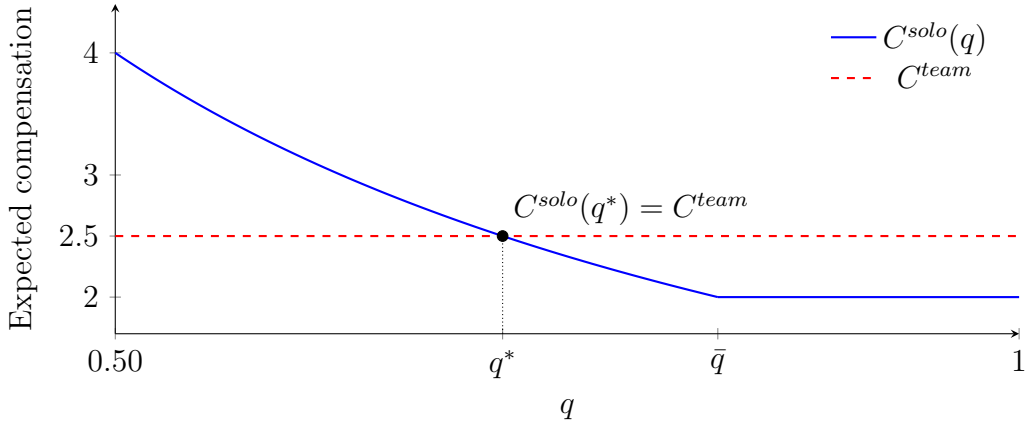


Figure 1: Expected compensation as q varies. The illustration sets $b = 1$, $\lambda_0 = 0.2$, and $\lambda_1 = 0.6$, so $q^* = 5/7$ and $\bar{q} = 5/6$. The team is cheaper to the left of q^*, while the solo operator is cheaper to its right.

The comparison has a simple interpretation. A solo operator avoids free-riding, but when q is low the principal must pay an information rent to prevent him from combining research

shirking with misreporting. A team eliminates this information rent, but splitting the tasks creates a fixed free-riding cost. The optimal organization trades off these two distortions. When lemons are easy to expose, the information problem is small and bundling dominates. When lemons are difficult to expose, the information rent becomes sufficiently large that separation is cheaper.

The cutoff in Proposition 1 has two useful features. First,

$$q^* < \bar{q}. \tag{15}$$

Thus, the organizational switch occurs while the solo operator's double-deviation constraint is binding. The team becomes attractive precisely because separation eliminates the information rent generated by that deviation, rather than because of the double-shirking constraint.

Second, the cutoff depends on $\theta = \lambda_1 - \lambda_0$, but not separately on λ_0, b, or the scale of compensation. A larger θ makes research more consequential for discovery quality. This exacerbates the researcher's free-riding problem under separation and therefore lowers q^*: when research has a larger effect on quality, the principal is willing to tolerate a less informative quality-verification technology before abandoning the solo operator.

The static benchmark thus isolates the central organizational tension. Bundling economizes on moral-hazard rents but creates an information rent; separation eliminates that information rent but creates free-riding. Whether two heads are better than one depends on which distortion is more costly.

Private information is essential to this comparison. If quality were publicly observable, the solo operator could be induced at cost $2b$, while the team's free-riding cost would remain unchanged, so bundling would always dominate. What the static model cannot explain, however, is why the preferred organization should change over the life of a project. In the dynamic model, the cost of controlling the solo operator's double deviation becomes history dependent, while the team's free-riding cost remains stationary. This history dependence generates endogenous transitions between organizational forms.

3 Dynamic Environment

We now extend the framework to a dynamic environment in which discoveries arrive over time, their quality depends on complementary tasks, and the principal must decide how to allocate these tasks as the project evolves.

3.1 Setting

Time is continuous. There are two tasks, which we continue to refer to as *research* and *development*. The development task requires a flow cost c and generates discoveries according to a Poisson process with intensity 1. The research task also requires cost c and improves the quality of each discovery as it arises. Without research, a discovery is a peach with probability $1-\theta$ and a lemon with probability θ; with research, all discoveries are peaches. Without loss of generality, we normalize the Poisson arrival rate to $\lambda = 1$ by rescaling time.

Each agent can perform either task or both tasks, provided the principal supplies the necessary resources. Agents are technologically identical, so differences in assignment reflect incentives rather than technological comparative advantage. An agent assigned only to development is the *developer*, an agent assigned only to research is the *researcher*, and an agent performing both tasks is a *solo innovator*.

At any time t, the agent(s) simultaneously choose their actions. The developer chooses $e_t \in A^D = \{s, w\}$, where s denotes shirking and w denotes working. The researcher chooses $a_t \in A^R = \{s_p, s_a, w\}$, where s_p denotes passive shirking, i.e., providing no research, and s_a denotes active shirking, providing limited input.[3] A solo innovator chooses the pair $(e_t, a_t) \in A^D \times A^R$.

Each action profile maps into an arrival rate of discoveries and a distribution over their quality. Shirking on development eliminates the possibility of making a discovery and yields private flow benefit b. Shirking on research affects the quality distribution of discoveries. If the researcher shirks passively, discoveries follow the baseline distribution without research. If the researcher shirks actively, lemons never arise, but the arrival rate of peaches is $(1-\theta) < 1$. Arrival rates of each type of discovery given any action profile appear in Table 1. For convenience, we assume that either type of shirking yields private benefit $b < c$. Allowing the private benefits to differ across types of shirking would not affect any qualitative results.

Development e_t	Research a_t	Peaches	Lemons
w	w	1	0
w	s_p	$(1-\theta)$	θ
w	s_a	$(1-\theta)$	0
s	$\cdots$	0	0

Table 1: Arrival rates under feasible action profiles.

As in the static benchmark, when a peach is built and accepted by the principal it yields the payoff $R > 0$. If a lemon is built and accepted by the principal, it develops a flaw with

[3] We discuss our motivation for allowing two forms of shirking in Section 6.

probability q and pays $-L < 0$. With probability $(1-q)$, no flaw develops and the lemon also yields R. Agency frictions arise because efforts and the quality of the discovery are both private to the agent. The principal observes when a product is discovered, but does not observe its quality. The principal only ex post observes the payoff R or $-L$ if the discovery is produced.

3.2 First Best

Before proceeding, it is useful to establish the first-best benchmark of the dynamic version of the model:

Lemma 3 *In the first best when the principal sees the agent's effort and the quality of any discovery, if $\theta > 1/2$, then the principal finances research and development until a peach is discovered. Her first best expected payoff is given by*

$$\pi = R - 2c. \tag{16}$$

Proof. With full information, the principal's options are: fund both tasks (peaches arrive at rate 1, yielding $R-2c$ in expectation), fund development only and discard lemons (peaches arrive at rate $1-\theta$, yielding $R-c/(1-\theta)$), or terminate. Producing lemons is never optimal since $(1-q)R - qL < 0$. Funding both tasks dominates development alone if and only if $2c \leq c/(1-\theta)$, i.e., $\theta \geq 1/2$. ■

Intuitively, research is worthwhile to the principal if it improves the likelihood of discovering a peach substantially. The lower bound for θ is $1/2$ in the first best but is higher when the principal is subject to asymmetric information, given the additional rents the agent must receive beyond those required to incentivize development alone. Because the question of whether one or two agents should be employed is only relevant when both tasks are desired, throughout the rest of paper we will impose a stronger condition to ensure that financing and incentivizing research is always preferred by the principal despite the agency cost:

Assumption 2 *Research is sufficiently valuable to the principal compared to its cost:*

$$\theta \;\geq\; 1 - \frac{c}{R} > \frac{1}{2}$$

In the Appendix (Section B.6), we derive a weaker sufficient condition that ensures research is always optimal. In what follows, we maintain this assumption so that the principal always invests in and incentivizes both tasks at all times until the project is terminated.

3.3 Contracts and Continuation Utilities

We index agents by $i \in \mathcal{I} = \{1, 2\}$ and tasks by $j \in \{R, D\}$. We work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The principal observes reported quality and realized payoffs; her information filtration is $\{\hat{\mathcal{F}}_t\}_{t\geq 0}$. Each agent i observes his own actions, any quality signals he receives, and his compensation; his filtration is $\{\mathcal{F}_t^i\}_{t\geq 0}$.

A contract specifies, for each agent i, an assignment process $\chi_{it} \in \{0, 1\}^2$ and a cumulative compensation process C_t^i, both adapted to $\hat{\mathcal{F}}_t$. The first component of χ_{it} indicates assignment to research, the second to development. Thus, $\chi_{it} = (1, 0)$ indicates that agent i is the researcher, $\chi_{it} = (0, 1)$ indicates that agent i is the developer, and $\chi_{it} = (1, 1)$ indicates that agent i is the solo innovator. Let $B(\chi_{it}, a_t, e_t)$ denote the flow of private benefits that accrue to agent i at time t under the assignment rule χ_t and action profile (a_t, e_t).

We refer to the date at which agent i is hired as $t_i = \inf\{t : \chi_{it} \neq (0, 0)\}$. Agent i is terminated at $\tau^i = \inf\{t : \chi_{is} = (0, 0) \text{ for all } s \geq t\}$. Let $\tau = \max\{\tau^1, \tau^2\}$ denote the project termination date.

Given a contract, agent i's continuation utility at time t is

$$W_t^i = \mathrm{E}\left[\int_t^{\tau^i} B(\chi_{is}, a_s, e_s)\, ds + \int_t^{\tau^i} dC_s^i \,\middle|\, \mathcal{F}_t^i\right],$$

and the principal's continuation value is

$$F_t = \mathrm{E}\left[Y_\tau - 2c(\tau - t) - \int_t^\tau (dC_s^1 + dC_s^2) \,\middle|\, \hat{\mathcal{F}}_t\right], \tag{17}$$

where $Y_\tau = R$ if a peach is produced, $Y_\tau \in \{R, -L\}$ if a lemon is produced, and $Y_\tau = 0$ if the project ends without a product. A contract is incentive compatible if each agent finds it optimal to exert effort on each task to which he is assigned and report quality truthfully. A contract is optimal if it maximizes the principal's date-0 payoff, F_0, over the set of contracts that are incentive compatible and deliver each agent non-negative initial utility: $W_{t_i}^i \geq 0$ for $i \in \{1, 2\}$.

4 The Agent(s) Problem

In this section we analyze the agent's IC constraints. First, we narrow the contracting space:

Lemma 4 *The optimal contract never induces shirking. It is without loss to defer all payments to date τ and the principal only develops a discovery that the agent reports to be a peach.*

The first part of the result arises because $b < c$, so any shirking can be strictly improved by skipping that portion of the contract and giving the agent compensation equal to his shirking benefits. The second part of the result arises because both the principal and the agents share the same discount factor (zero) so deferring payments to when the contract ends is costless. Furthermore, because the agent is risk-neutral, any payment that the agent would receive when the contract ends without a product can be moved to the state in which a product is generated. Finally, because $(1-q)R < qL$, the principal never accepts a lemon if the agent reports so.

Let N_t denote the counting process of successful discoveries, adapted to $\hat{\mathcal{F}}_t$. For any incentive compatible contract, there exists a predictable process β_t^i, adapted to $\hat{\mathcal{F}}_t$, such that

$$dW_t^i = \beta_t^i(dN_t - dt), \tag{18}$$

The evolution of the agent's continuation utility given in (18) is based on the standard martingale representation theorem and the fact that the agent does not discount future payoffs. Intuitively, β_t^i is the jump in W_t^i when a peach is produced. In the absence of discoveries, W_t^i drifts downward at rate β_t^i until it hits zero, at which point the agent is terminated.

Proposition 2 *Given the assignment χ_{it}, effort is incentive compatible if and only if*

$$\beta_t^i \;\geq\; \begin{cases} \beta_D \equiv b, & \chi_{it} = (0,1) \text{ (developer)}, \\ \beta_R \equiv b/\theta, & \chi_{it} = (1,0) \text{ (researcher)}, \\ \beta_S(W_t^i) \equiv \max\{2b,\ b/(\theta q) - W_t^i\}, & \chi_{it} = (1,1) \text{ (solo innovator)}, \end{cases}$$

and all inequalities bind in the optimal contract.

Below, we derive the IC conditions in Proposition 2 given the agent's assignment, and elaborate their implications on the cost of hiring one versus two agents.

4.1 Solo Innovator

To incentivize the solo innovator, the principal must guard against a set of possible deviations. The agent can shirk on either or both tasks and the agent can misreport lemons as peaches or vice versa. We can split the incentive compatibility condition into two sets of constraints. The first set of constraints ensures that the agent exerts effort on both tasks conditional on truthful reporting. The second set of constraints ensures that the agent reports the quality of discoveries truthfully.

Let β_t^S denote the solo innovator's reward for reporting a peach that generates R and let γ_t^S denote the reward for reporting a lemon. The agent's expected flow benefit from exerting effort on both tasks is β_t^S. Consider first the strategy of shirking on both tasks (*double shirking*), which generates flow private benefit $2b$ and no possibility of a discovery. Preventing this deviation requires $\beta_t^S \geq 2b$. If this constraint is satisfied, we can immediately rule out two other deviations: (1) shirking only on development, which also yields no possibility of discoveries but a private benefit of only b, and (2) working on development and a-shirking on research, which yields a flow benefit of $b + (1-\theta)\beta_t^S \leq \beta_t^S$ since $\theta \geq 1/2$. Therefore, the effort IC can be reduced to:

$$\beta_t^S \geq \max\left\{ \underbrace{2b}_{\text{double shirking}}, \underbrace{(1-\theta)\beta_t^S + \theta\gamma_t^S + b}_{\text{p-shirking}} \right\}. \tag{IC Effort}$$

Next, consider the solo innovator's incentive to report truthfully. Reporting peaches truthfully is incentive compatible provided that $\beta_t^S \geq \gamma_t^S$, which as we will see, is not a binding constraint. If the agent misreports a lemon as a peach, then the principal develops the discovery. With probability $1-q$, no flaw materializes, the principal realizes R, and the agent is rewarded β_t^S. With probability q, a flaw materializes, which reveals that the agent lied and the principal punishes the agent to the maximum extent possible. However, due to limited liability, the maximum punishment is simply to terminate the agent. Hence, truthfully reporting lemons requires

$$\gamma_t^S + W_t^S \geq (1-q)(\beta_t^S + W_t^S) + q \cdot 0 \tag{19}$$

Rearranging terms, we get

$$\gamma_t^S \geq (1-q)\beta_t^S - qW_t^S. \tag{IC Report}$$

Observe that making either constraint (IC Effort) or (IC Report) slack will make the other constraint harder to satisfy. Therefore, both constraints will hold with equality at the optimum. Substituting the value for γ_t^S from (IC Report) into (IC Effort), we arrive at the expression in Proposition 2:

$$\beta_t^S = \beta_S(W_t^S) = \max\left\{2b, \frac{b}{\theta q} - W_t^S\right\}. \tag{20}$$

When W_t^S is high, the agent has substantial "skin in the game," so the threat of termination is powerful and the binding constraint is double shirking. When W_t^i is low, the agent is

more tempted to misreport a lemon as a peach, so truthful reporting requires $\gamma_t^S > 0$, which in turn raises the attractiveness of p-shirking.[4] Whether this temptation exceeds the benefit of double shirking depends on model parameters. Define

$$K \equiv \frac{b}{\theta q} \qquad \text{and} \qquad \underline{W} \equiv K - 2b = \frac{b(1-2\theta q)}{\theta q}, \tag{21}$$

so that $\beta_S(W) = K - W$ for $W < \underline{W}$ and $\beta_S(W) = 2b$ for $W \geq \underline{W}$. If $\theta q \geq 1/2$, then $\underline{W} \leq 0$ and $\beta_S(W) = 2b$ for all $W \geq 0$.

Recall that q corresponds to the likelihood that a lemon turns out to be flawed. The lower q is, the more difficult it is for the principal to distinguish the lemon from the peach *ex post* and punish the agent for lying. Similar to when the agent has a lower continuation value, this makes the double deviation more tempting, which requires stronger incentives to dissuade. $\underline{W}$ captures the continuation value at which the agent is indifferent between double shirking and the double deviation. The principal's cost of providing incentives is constant when $W_t^S > \underline{W}$ and increases as W_t^S declines below $\underline{W}$, reaching the highest level as $W_t^S \to 0$. Notice that $\underline{W}$ decreases with q. When q is sufficiently high, $\underline{W} \leq 0$. In this case, the principal only needs to compensate the agent for the private benefits from double shirking so β_t^S is always constant ($2b$).

4.2 Two Agents

We now derive the IC conditions when the two tasks are assigned to separate agents. The timing of the stage game is as follows. Each agent simultaneously and privately chooses whether to put in effort. If a discovery is made, the principal solicits a report of its quality. As in the case of the static benchmark. The principal achieves the minimum cost if the report comes from the developer. If the developer reports a peach, the principal accepts the discovery and realizes R or $-L$. If he reports a lemon, the discovery is discarded. We require the equilibrium of the stage game to be subgame perfect.

Suppose the researcher exerts effort. For the developer, the expected flow benefit from working is β_t^D. If he shirks, he receives private flow benefit b and nothing is produced. Thus, incentivizing the developer to work requires:

$$\beta_t^D \geq b. \tag{22}$$

[4] Producing lemons requires the agent's effort in development, so the agent cannot reap unlimited utility by continuously fabricating lemons.

To ensure the developer truthfully reports peaches and lemons requires

$$\beta_t^D \geq \gamma_t^D, \tag{23}$$

$$\gamma_t^D \geq (1-q)\beta_t^D - qW_t^D. \tag{24}$$

In equilibrium, the researcher will exert effort and thus lemons will not be produced on path. Therefore, γ_t^D does not enter the developer's IC constraint, which means the principal does not have to give information rents to the developer for reporting truthfully. Critically, this means the principal has no need to elicit information from the researcher and can use the developer's report to provide incentives to the researcher, as that in the static benchmark.

The IC condition for the researcher is

$$\beta_t^R \geq \max\{\underbrace{(1-\theta)\beta_t^R + b}_{\text{a-shirking}}, \underbrace{(1-\theta)\beta_t^R + b + \theta\gamma_t^R}_{\text{p-shirking}}\} \tag{25}$$

The principal can set $\gamma_t^R = -W_t^R$ to deter p-shirking. Importantly, as long as $\gamma_t^R \leq 0$, which is always feasible, a-shirking dominates p-shirking. That is, given that the developer will report lemons truthfully, the researcher cannot gain from generating lemons via p-shirking. Thus, if he decides to shirk on research, he is always better off shirking actively so as not to generate lemons. Hence, the a-shirking constraint will bind under the optimal contract and $\beta_t^R = \beta_R = b/\theta > b$. The last inequality is due to the same free-riding problem in the static benchmark, which reflects the additional cost of the two-agent team compared to the minimum possible cost of incentivizing a solo operator.

4.3 Cost Comparison.

Comparing the IC conditions for two agents and those for a solo innovator reveals the following result regarding the cost of either organizational structure:

Proposition 3 *The principal's flow cost of incentivizing two separate agents is lower than the flow cost of a solo innovator if both q and the continuation utility of the solo innovator are sufficiently low, i.e.,*

$$\beta_D + \beta_R < \beta_S(W_t^S) \text{ if } q < \frac{1}{1+\theta} \text{ and } W_t^S < K - \frac{b(1+\theta)}{\theta} \tag{26}$$

Notice that the condition $q < \frac{1}{1+\theta}$ is identical to the condition for two agents to be cheaper in Proposition 1 as they both capture the incremental likelihood of peach if the research effort is spent. Thus, the intuition for why two agents can be cheaper than a solo innovator from

the static model still applies. The two-agent team creates a free-riding problem but saves the cost of inducing truthful reporting, where the latter is higher when q is lower.

Yet, the dynamic model also illustrates an additional insight: the cost of a solo innovator is *time-varying* depending on how much continuation utility, or time, the solo innovator has left. The intuition of the static model is a special case for exactly *one attempt* at developing a high quality product, which corresponds to when $W_t^S \to 0$. The cost of incentives for the solo innovator is lower as the "number of attempts" grows, while the total cost of the team is constant.

In summary, the organizational structure determines which deviations are binding and, therefore, the total cost of incentive provision. By consolidating the two tasks to a solo innovator, the principal can leverage economies of scope and offer a lower total reward to motivate effort (i.e., $2b < b + \frac{b}{\theta}$). However, a solo innovator must be disciplined against the double deviation, which is more expensive when the agent's continuation value is low and when lemons are difficult to detect.

5 Optimal Contract

Having characterized the incentive compatibility conditions under each organizational regime and explored the trade-offs between a solo innovator and a two-agent team, we are now ready to address our primary question of interest: what is the optimal dynamic arrangement that maximizes the principal's expected payoff at the outset of the contract? Because the two agents are identical, we will label the agent that is hired first as agent 1, and the second as agent 2. If both agents are hired at the same time, agent 1 refers to the agent initially assigned to development. Before proceeding, we impose a parametric assumption that ensures the principal would consider using a two-agent team.

Assumption 3

$$\pi > \frac{(1+\theta)b}{\theta} = \beta_D + \beta_R \tag{27}$$

This ensures that the expected output exceeds the cost of providing incentives to the team for one "instant". Without this assumption, the principal would never use a structure with a two-agent team. We do not make the analogous assumption for a structure with a solo innovator because it depends on our parameter of interest, q, which also plays an important role in the determination of the optimal organizational design.

5.1 When Both Are Hired

We start off by solving the task assignment problem *given* that both agents have been hired with exogenous continuation values W_1 for agent 1 and W_2 for agent 2. Let $V(W_1, W_2)$ denote the principal's value function given the continuation value of each agent. Thus, for each $\vec{W} = (W_1, W_2) \in \mathbb{R}^2_+$ the principal has four options:

1. Use Agent 1 as a solo innovator.
2. Use Agent 2 as a solo innovator.
3. Use Agent 1 as a researcher and Agent 2 as a developer.
4. Use Agent 1 as a developer and Agent 2 as a researcher.

The HJB for the principal's problem is

$$V(W_1, W_2) = \pi - (W_1 + W_2) - \min \left\{ \begin{array}{c} \beta_S(W_1)(1 + V_1), \\ \beta_S(W_2)(1 + V_2), \\ \beta_D(1 + V_1) + \beta_R(1 + V_2), \\ \beta_R(1 + V_1) + \beta_D(1 + V_2) \end{array} \right\} \tag{28}$$

where $V_i = \frac{\partial V}{\partial W_i}$. The boundary conditions for the principal's problem are

$$V(W_1, 0) = F_1(W_1)$$
$$V(0, W_2) = F_1(W_2)$$

where F_1 is the principal's value function with a single agent assigned to both tasks (see Appendix A for derivation).

Because the problem is symmetric, it suffices to focus on either the region where $W_1 \geq W_2$ or where $W_2 \geq W_1$. We focus on the latter. As will be shown later, this is also where the equilibrium path lies (recall agent 1 is the first agent assigned to one or more tasks). Define $\mathcal{W}_2 = \{\vec{W} \in \mathbb{R}^2_+ : W_2 \geq W_1\}$.

Because V is symmetric and must be weakly concave in both of its arguments,[5] it will never be optimal to assign the agent with less continuation value to both tasks or to the more expensive task. Intuitively, it is better to assign the agent with the higher continuation value to both tasks or the task that requires stronger incentives. Hence, for $W_2 \geq W_1$, we

[5]Symmetry follows because the agents are interchangeable: relabeling the agents in any incentive-compatible contract yields an incentive-compatible contract with the same cost. Concavity follows because the principal can randomize over continuation values, which convexifies the set of attainable payoffs.

only need to consider two of the four possible allocations: assign agent 2 to both tasks or assign agent 2 to research and assign agent 1 to development. The HJB for the principal's problem can be simplified to

$$V(\vec{W}) = \pi - (W_1 + W_2) - \min\left\{\beta_S(W_2)(1 + V_2(\vec{W})), \beta_R(1 + V_2(\vec{W})) + \beta_D(1 + V_1(\vec{W}))\right\}.$$

Define W_S such that $\beta_S(W_S) = \beta_D + \beta_R$, which is the continuation value such that assigning one agent to both tasks requires the same reward as using one agent for each task. Direct calculation yields

$$W_S = \frac{b(1 - q(1 + \theta))}{\theta q} \tag{29}$$

which is non-negative if $q \leq 1/(1 + \theta)$. The *myopic solution*, which minimizes the total reward to both agents in each state, is to assign agent 2 to both tasks when $W_2 > W_S$ and each agent to one task when $W_2 < W_S$. The optimal task allocation differs from the myopic solution because it accounts for the influence the current assignment has on the cost of future assignments.

To understand where this difference arises, consider the point labeled A in Figure 2. Starting from point A, the myopic solution assigns agent 1 to development and agent 2 to research. In this case, both agents' continuation values decrease if the project does not succeed. Because agent 2 is assigned to the more expensive task, her continuation value decreases faster than agent 1's, i.e., $dW_2/dW_1 = \beta_R/\beta_D = \frac{1}{\theta}$. Hence, $\vec{W}_t$ travels downward along the line with slope $1/\theta$ until reaching the y-axis at which point agent 1 is terminated and agent 2 is assigned to both tasks.

In contrast, the optimal solution starting from point A assigns agent 2 to both tasks, so that $\vec{W}_t$ travels vertically downward until reaching $\mathcal{K}$, at which point each agent is assigned to a single task. In both the optimal and myopic solutions, the amount of time that each agent is assigned to one task is the same. The difference is that under the optimal solution, agent 2 is assigned to both tasks when her continuation value is relatively high, and hence when incentive costs are relatively low. As a result, the principal can increase the time agent 2 works on both tasks by front-loading this assignment, rather than postponing it until agent 1 is terminated, as in the myopic solution.

We now characterize the solution to the task assignment program. In order to do so, let $\mathcal{K} = \{\vec{W} \in [0, W_S]^2 : \theta W_1 \leq W_2 \leq W_1/\theta\}$ denote the "kite" shaped region that extends northeast from the origin. Let $\mathcal{A}_i = \{\vec{W} : W_i > \min\{W_j/\theta, W_S\}, W_j < W_S\}$ and let $\mathcal{E} = \{\vec{W} : W_1, W_2 > W_S\}$.

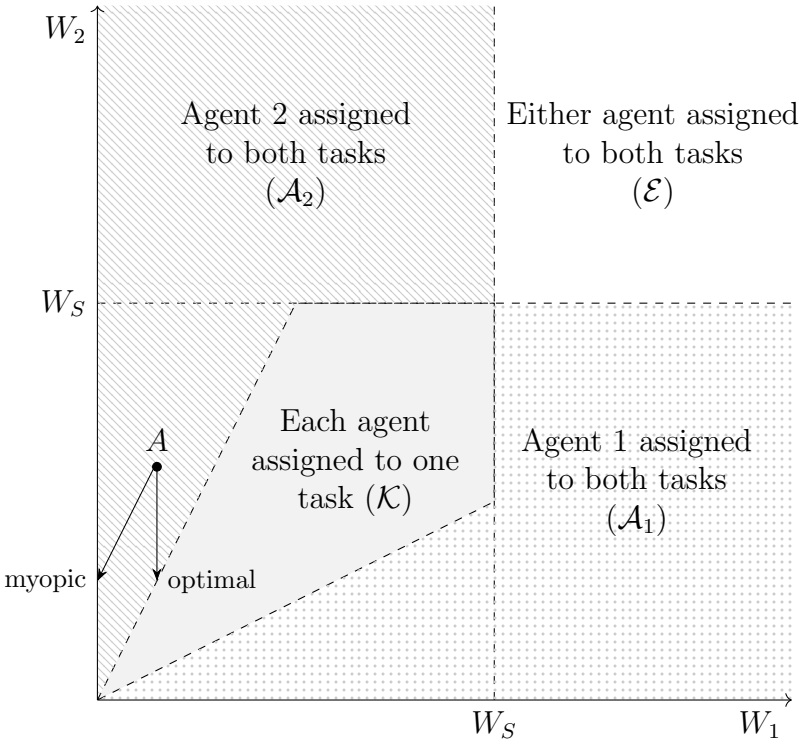


Figure 2: Illustration of the solution to the task assignment problem

Proposition 4 *The solution to the task assignment problem is as follows:*

- *If $\vec{W} \in \mathcal{K}$, it is optimal to assign one task to each agent.*
- *If $\vec{W} \in \mathcal{A}_i$, then it is optimal to assign both tasks to agent i.*
- *If $\vec{W} \in \mathcal{E}$, then allocating both tasks to either agent is optimal.*

5.2 When to Hire the Second Agent

Having characterized the solution to the task assignment problem and the principal's value function $V(W_1, W_2)$ once both agents are hired, we now determine *when* it is optimal for the principal to bring in the second agent. At this stage of the project, assume that only agent 1 is active and the principal can decide when to hire agent 2. The decision problem thus takes the form of a stopping problem: continue with a single solo innovator or hire the second agent.

5.2.1 The Principal's Problem

If the principal hires the second agent when the state reaches $W_t = w$, she enters the two-agent regime characterized by Proposition 4. Conditional on hiring at that point, she chooses

the initial promise to the new agent to maximize her continuation payoff:

$$G(w) \;=\; \max_{y \geq 0} V(w, y).$$

The function $G(w)$ gives the principal's value *from hiring agent 2 immediately* when agent 1's continuation value is w. Because the principal can randomize over continuation utilities for agent 1 at the hiring date, the relevant object for the stopping problem is the *concave envelope* of G,

$$\bar{G}(w) \;=\; \sup\{\, g \mid (w, g) \in \mathrm{co}(G) \,\},$$

which represents the best payoff attainable from possibly mixing over agent's continuation values at the time of hire.[6] We refer to $\bar{G}$ as the *obstacle*.

Starting with a single agent, the principal chooses the initial promise W_0 and a stopping time τ at which to hire the second agent to maximize her payoff:

$$\Pi = \sup_{(\tau, W_0)} \int_0^\tau e^{-t}\big(\pi - W_t - \beta_S(W_t)\big)\, dt + e^{-\tau}\, \bar{G}(W_\tau), \qquad dW_t = -\beta_S(W_t)\, dt. \tag{30}$$

The first term is the expected payoff from continuing with one agent until τ, while the second term is the continuation value if agent 2 is hired at that point. Let (τ^*, W_0^*) denote the solution to the problem and let $\hat{W}_1 = W_{\tau^*}$.

Differentiating (30) with respect to τ gives the marginal value of waiting an additional instant before hiring the second agent:

$$\frac{d\Pi}{d\tau} = e^{-\tau}\, \Phi(W_\tau), \qquad \Phi(W) \equiv \pi - W - \beta_S(W)\big(1 + \bar{G}'(W)\big) - \bar{G}(W).$$

The function $\Phi(W)$ therefore captures the *benefit of delay*: if $\Phi(W) > 0$, it is optimal to wait longer before hiring agent 2, while if $\Phi(W) < 0$, the principal should hire agent 2 immediately. An interior solution $\hat{W}_1$ must satisfy the first-order condition

$$\Phi(\hat{W}_1) = 0, \tag{31}$$

which corresponds to a smooth-pasting condition between the single-agent and two-agent

[6] For any point $(w, g) \in co(G)$, there exists a distribution μ over continuation values for agent 1 such that $E_\mu W_1 = w$ and $E_\mu G(W_1) = g$. Therefore, conditional on hiring agent 2 when agent 1's continuation value is W_1, the principal can achieve a payoff of $\bar{G}(W_1)$.

value functions. Observe that for all $W \in [0, W_S]$,

$$\Phi'(W) = -(K - W)\bar{G}''(W) \geq 0.$$

Thus, Φ satisfies the (weak) single-crossing property.

Connection to the QVI formulation. The stopping problem in (30) is equivalent to a oneobstacle quasi-variational inequality (QVI) for the principal's value function $F(W)$ when she has the option to hire a second agent. Let the solo innovator $\mathcal{A}$ act on F as

$$\mathcal{A}F(W) = (\pi - W - \beta_S(W)) - \beta_S(W)F'(W) - F(W)$$

The principal's value function then satisfies

$$\max\left\{\mathcal{A}F(W),\ \bar{G}(W) - F(W)\right\} = 0, \qquad F(0) = \bar{G}(0). \tag{32}$$

In the continuation region ($W > \hat{W}_1$) the first term binds, while in the switching region ($W \leq \hat{W}_1$) we have $F(W) = \bar{G}(W)$. At an interior threshold $\hat{W}_1$, the free-boundary conditions

$$F(\hat{W}_1) = \bar{G}(\hat{W}_1), \qquad F'(\hat{W}_1) = \bar{G}'(\hat{W}_1), \qquad \mathcal{A}F(\hat{W}_1) = 0$$

are value matching, smooth pasting, and the continuation HJB at the boundary, respectively. Substituting value matching and smooth pasting into the continuation HJB yields the first-order condition $\Phi(\hat{W}_1) = 0$ defined above. Hence, the QVI yields the solution to the stopping problem for any initial W_0: $\Phi(W)$ is simply the HJB operator evaluated at the obstacle $\bar{G}$. Once we have solved (32) for F, we can then turn to the choice of the optimal W_0^*.

We proceed with two useful lemmas and then state our main result.

Lemma 5 *Let $W_2^*(w) \in \arg\max_{y \geq 0} V(w, y)$. If $(w, W_2^*(w)) \in \mathcal{A}_1 \cup \mathcal{E}$ then $\Phi(w) \geq 0$. In words: if, conditional on hiring agent 2, it is optimal to assign agent 1 to both tasks, then it is optimal to wait to hire agent 2.*

The lemma simply says there is no reason to "hire" agent 2 unless it is strictly optimal for the principal to put agent 2 to work conditional on being hired. This result is intuitive. Nevertheless, it is useful in restricting the domain over which it is necessary to look for an interior solution. Let $\mathcal{W}_1 = \{w \in \mathbb{R}_+ : (w, W_2^*(w)) \in \mathcal{K} \cup \mathcal{A}_2\}$ denote the relevant domain.

Next, we decompose $\mathcal{W}_1$ into two sub-regions: the stopping region ($\mathcal{S}$) and the continuation region ($\mathcal{C}$). We include K as an argument for each sub-region in order to highlight the

sub-regions dependence on our key parameter of interest.

$$\mathcal{C}(K) \equiv \{W \in \mathcal{W}_1 : \Phi(W; K) > 0\}$$
$$\mathcal{S}(K) \equiv \{W \in \mathcal{W}_1 : \Phi(W; K) \leq 0\}$$

In words, $\mathcal{C}(K)$ is the region where the marginal benefit of delay is positive and therefore it is optimal for the principal to delay hiring agent 2. Whereas $\mathcal{S}(K)$ is the stopping region where immediately hiring agent 2 is optimal. Since Φ is weakly increasing in W, both $\mathcal{C}$ and $\mathcal{S}$ are intervals, which can be characterized by $\hat{W}_1(K)$, which is the largest value in $\mathcal{W}_1$ such that $\Phi(\hat{W}_1) \leq 0$. We let $\hat{W}_1(K) = 0$ if Φ is strictly positive on $\mathcal{W}_1$. Thus, $\mathcal{S}(K) = [0, \hat{W}_1(K)]$ and $\mathcal{C}(K) = (\hat{W}_1(K), \max(\mathcal{W}_1)]$.

Lemma 6 *There exists positive and finite $K_{\text{seq}} < K_{\text{sim}}$ such that*

1. *For all $K \leq K_{\text{seq}}$, $\hat{W}_1(K) = 0$.*
2. *For all $K \in (K_{\text{seq}}, K_{\text{sim}})$, $\hat{W}_1(K) \in (0, \max(\mathcal{W}_1))$.*
3. *For all $K \geq K_{\text{sim}}$, $\hat{W}_1(K) = \max(\mathcal{W}_1)$.*

Hiring the second agent involves making a fixed cost investment (i.e., $W_2^*(W_1)$) that the principal would like to avoid. However, as W_1 decreases, it becomes more expensive to deter the double deviation. For intermediate K, the principal resolves the tradeoff by waiting until W_1 reaches the critical level $\hat{W}_1(K)$ before adding the second agent to the team. When K is small, the cost of deterring the double deviation is low, and the optimal policy is to wait until $W_1 = 0$ (and agent 1 is terminated) before hiring agent 2. When K is large, the double deviation is very costly to deter, making early hiring of the second agent optimal despite the fixed cost.

5.3 Organizational Regimes

Given any initial utility for Agent 1, Lemma 6 characterizes the optimal policy for when to hire the second agent $\tau = \inf\{t : W_t \leq \hat{W}_1(K)\}$. All that remains is to determine the optimal initial utility for the first agent. There are two possibilities. Either $W_0^* > \hat{W}_1$ in which case agent 1 works as a solo innovator before agent 2 is hired, or $W_0^* = \hat{W}_1$, in which case the two agents are hired simultaneously. In this latter case, the initial starting point is not uniquely determined because the principal's value function is flat in the interior of the kite region. However, if there is an arbitrarily small cost of reallocating tasks to agents, then

the solution is unique (up to the choice between two symmetric optimal paths). In what follows, we apply this selection criterion.

If $W_0^* > \hat{W}_1$, then the principal hires agent 1 before agent 2. Effectively, the principal "attaches" the single agent value function to $\bar{G}$. Hence, the principal's value function is linear on $[\hat{W}_1, \underline{W}]$ with slope A that is determined by the boundary condition

$$F(\hat{W}_1) = A\hat{W}_1 + \pi - (1 + A)K = \bar{G}(\hat{W}_1)$$

Rearranging terms, we see that

$$A(\hat{W}_1) = \frac{\pi - K - \bar{G}(\hat{W}_1)}{K - \hat{W}_1} \tag{33}$$

For intuition, we can note that $A(\hat{W}_1)$ corresponds to $d\Pi/dW_0$ evaluated at $W_0 = \hat{W}_1$. Thus, if $A(\hat{W}_1)$ is positive (negative), then the solution must involve $W_0^* > \hat{W}_1$ ($W_0^* = \hat{W}_1$). Note also that at an interior threshold, the first-order condition $\Phi(\hat{W}_1) = 0$ rearranges to exactly $A(\hat{W}_1) = \bar{G}'(\hat{W}_1)$: the slope of the attached single-agent value function equals the slope of the obstacle at the contact point (smooth pasting).

Lemma 7 *Agent 1 is hired strictly before Agent 2 if and only if $K < K_{\text{sim}}$.*

From Lemmas 6 and 7, we can conclude that the optimal organizational design can take one of three possible forms as summarized in the following proposition.

Proposition 5 (Organizational Regimes) *The optimal organizational regime is as follows:*

1. ***Sequential Generalist:*** *For $K \leq K_{\text{seq}}$: the project starts with agent 1 as the solo innovator. The principal hires agent 2 when agent 1 is terminated. Each agent is assigned to both tasks during their employment.*

2. ***Simultaneous Specialist:*** *For $K \geq K_{\text{sim}}$, the principal hires both agents at the same time. Each agent is assigned to one task during their employment. Both agents are terminated at the same time.*

3. ***Staggered Overlap:*** *For $K \in (K_{\text{seq}}, K_{\text{sim}})$, the project starts with agent 1 as the solo innovator. Agent 2 is hired before agent 1 is terminated.*

In the sequential-generalist regime, it is optimal for a single agent to complete both tasks; effectively, whichever agent is employed acts as the solo innovator. By contrast, in the

simultaneous-specialist regime, two heads are preferred to one, and each agent is assigned to a distinct task for the entire duration of the project. Importantly, these "specialists" do not arise from inherent differences in ability, but rather from the principal's need to control agency rents. The research task is more costly to incentivize due to the free-rider problem, so the agent assigned to research must be granted higher rents per unit of time than the developer. While the principal could, in principle, equalize the agents' initial utilities, assign agent 1 to research and agent 2 to development, and then rotate their roles midway through the project, doing so would require incurring (albeit small) reallocation costs that she prefers to avoid. The optimal policy is therefore to grant one agent a higher initial utility and have him specialize in research for the duration of the project.

In the staggered overlap structure, agent 1 is assigned to both tasks for a period of time before agent 2 is hired. If agent 1 does not succeed by the time her continuation value reaches $\hat{W}_1$, then agent 2 is hired. How tasks are allocated to agents once agent 2 is hired can subsequently take one of two forms.

1. *Standby Developer*: This occurs when $(\hat{W}_1, W_2^*(\hat{W}_1)) \in \mathcal{A}_2$. In this case, agent 2 takes over both tasks upon being hired, while agent 1 is put on "standby." If agent 2 does not succeed for a while then eventually, the state reaches the kite region. At that point, agent 1 takes over development, while agent 2 continues research.

2. *External Consultant*: This occurs when $(\hat{W}_1, W_2^*(\hat{W}_1)) \in \mathcal{K}$. In this case, agent 2 is brought in as an "external consultant" in charge of research for the remaining duration of the project, while agent 1 continues to develop.

Formally, define

$$\tilde{K} \equiv \sup\left\{K \in (K_{\text{seq}}, K_{\text{sim}}) : \ (\hat{W}_1(K), W_2^*(\hat{W}_1(K))) \in \mathcal{A}_2\right\}, \tag{34}$$

with the convention $\sup \emptyset = K_{\text{seq}}$. As shown in the Appendix (proof of Proposition 6), $\hat{W}_1(K)$ is weakly increasing in K, so the staggered region splits into at most two intervals.

Proposition 6 *The threshold $\tilde{K}$ satisfies $\tilde{K} < K_{\text{sim}}$. Moreover:*

1. *If $K_{\text{seq}} < \tilde{K} < K_{\text{sim}}$, then the Standby Developer structure is optimal for all $K \in (K_{\text{seq}}, \tilde{K})$, and the External Consultant structure is optimal for all $K \in (\tilde{K}, K_{\text{sim}})$.*

2. *If $\tilde{K} = K_{\text{seq}}$, then the External Consultant structure is optimal for all $K \in (K_{\text{seq}}, K_{\text{sim}})$.*

In the Appendix, we provide a necessary and sufficient condition for $\tilde{K}$ to be interior (Condition 1: standby capacity must be valuable at the margin, and no lottery over the incumbent's standby stake may dominate a small standby promise.

To provide intuition, we leverage monotonicity of the solution with respect to K. At the boundaries:

- When $K \leq K_{\text{seq}}$: $\hat{W}_1(K) = 0$, so agent 2 is hired only after agent 1 is terminated (sequential hiring).
- When $K \geq K_{\text{sim}}$: $\hat{W}_1(K) = W_0^*$, so both agents are hired simultaneously.

For intermediate values $K \in (K_{\text{seq}}, K_{\text{sim}})$, the hiring threshold $\hat{W}_1(K)$ increases weakly between these extremes (jumping only where $\bar{G}$ has a linear segment).

When $\tilde{K} > K_{\text{seq}}$, for K just above K_{seq} the hiring threshold $\hat{W}_1(K)$ is close to zero—agent 1 has little continuation value remaining when agent 2 is hired. At this point, agent 1 is expensive to incentivize, while the freshly hired agent 2 has high continuation value and is cheap to incentivize. The optimal response is to hire agent 2 in $\mathcal{A}_2$ and let her operate as a solo innovator for a while. Agent 1's remaining continuation value is *preserved* for when it is most valuable: if agent 2 does not succeed and the state eventually enters the kite region $\mathcal{K}$. At that point, agent 1 rejoins to handle development while agent 2 continues research.

For K closer to K_{sim}, the hiring threshold $\hat{W}_1(K)$ is larger—agent 1 still has substantial continuation value when agent 2 is hired. Both agents now have enough skin in the game that splitting tasks is immediately optimal. Agent 2 enters as an "external consultant" for research, while agent 1 continues development. The agents work in parallel from the outset. Whenever $\hat{W}_1(K)$ passes through the boundary between $\mathcal{A}_2$ and $\mathcal{K}$ continuously, $\tilde{K}$ is precisely the value of K at which the optimal hiring point $(\hat{W}_1(K), W_2^*)$ lies on that boundary. If instead $\hat{W}_1(K)$ jumps across the boundary at K_{seq}, then $\tilde{K} = K_{\text{seq}}$ and the Standby Developer structure does not materialize.

6 Discussion of Assumptions and Extensions

In this section, we discuss several of our model assumptions and the implications of relaxing them.

6.1 Lemons on Path

When both agents exert effort, lemons are not produced on path. This assumption is convenient and simplifies the analysis because the developer's bump in utility from reporting a

lemon does not materialize. Suppose instead that lemons are produced on path with probability $\epsilon > 0$, either when both agents exert effort or when the developer exerts effort and the researcher actively shirks. The arrival rate of lemons and peaches in this extension of the model is:

Development e_t	Research a_t	Peaches	Lemons
w	w	$(1-\epsilon)$	ϵ
w	s_p	$(1-\theta)$	θ
w	s_a	$(1-\theta-\epsilon)$	ϵ
s	$\cdots$	0	0

Table 2: Arrival rates under feasible action profiles with lemons on path.

In this case, the developer's continuation value increases following the report of a lemon, which makes characterizing the full solution to the principal's problem more involved. But the underlying economics remain unchanged.

Proposition 7 *Suppose lemons arise on path as in Table 2 and fix $\epsilon \in (0, \theta)$. Then the following contracts implement effort on both tasks and truthful reporting, and are optimal within their respective organizational regimes.*

1. *Solo Innovator:*

$$\beta_\epsilon^S(W^S) = \max\{K_\epsilon - W^S,\ \bar{\beta}_\epsilon\}, \qquad \gamma_\epsilon^S(W^S) = \max\{(1-q)K_\epsilon - W^S,\ \bar{\gamma}_\epsilon\},$$

where $K_\epsilon := \frac{b}{q(\theta-\epsilon)}$, $\bar{\beta}_\epsilon := 2b + \epsilon\frac{b}{\theta-\epsilon}$, $\bar{\gamma}_\epsilon := 2b - \frac{(1-\epsilon)b}{\theta-\epsilon}$.

2. *Two-Agent Team:*

$$\beta_\epsilon^D = \gamma_\epsilon^D = b, \qquad \beta_\epsilon^R = \frac{b}{\theta-\epsilon}, \qquad \gamma_\epsilon^R = 0.$$

This structure closely mirrors Proposition 2 (the benchmark with $\epsilon = 0$). A new feature when lemons occur on path is that the principal may optimally reward the solo innovator for lemons even when the agent's continuation value is high. In particular,

$$\bar{\gamma}_\epsilon > 0 \quad \Longleftrightarrow \quad \theta > \frac{1+\epsilon}{2}.$$

The intuition comes from the double-shirking constraint,

$$(1-\epsilon)\beta + \epsilon\gamma \geq 2b,$$

which requires the principal to provide sufficient incentives in expectation across peach and lemon realizations. While the principal would like to keep both instruments small, β is paid as a transfer upon a produced peach whereas γ can be delivered by increasing promised continuation utility following a reported lemon. Shifting compensation weight from β toward γ reduces the principal's expected transfer burden, making γ a relatively cheap way to satisfy the double-shirking constraint when lemons arrive with positive probability. This substitution is limited by the other IC constraints – most importantly, passive shirking and truthful reporting – which limit the extent to which the principal can rely on rewarding lemons to provide incentives.

Given that agents are now compensated for both peaches and lemons, comparing the total reward for peaches alone (as we did in Proposition 3) is no longer sufficient to characterize the total incentive cost of each organizational structure. Nevertheless, we can still provide an analogous and equally sharp characterization of when each regime is more cost-effective.

Corollary 1 *Define*

$$\bar{W}_\epsilon \;:=\; (1-q)K_\epsilon - b \;=\; \frac{(1-q)b}{q(\theta-\epsilon)} - b,$$

which is decreasing in q and strictly positive if and only if $q < \frac{1}{1+\theta-\epsilon}$. The principal's flow cost of incentivizing a solo innovator is lower than that of the two-agent team, componentwise, if and only if $W^S > \bar{W}_\epsilon$, i.e.,

$$\beta_\epsilon^S(W^S) < \beta_\epsilon^D + \beta_\epsilon^R \;\; \textit{and} \;\; \gamma_\epsilon^S(W^S) < \gamma_\epsilon^D + \gamma_\epsilon^R \quad \Longleftrightarrow \quad W^S > \bar{W}_\epsilon.$$

Both inequalities reverse if $W^S < \bar{W}_\epsilon$.

6.2 Forms of Shirking on Research

We allow two forms of shirking for the researcher in the dynamic model. To see why, consider an alternative version of the model in which the researcher can only shirk passively. In this case, the researcher cannot obtain private benefits without producing lemons. The principal can then use the developer's report of a lemon to induce the researcher to exert effort. In particular, by setting $\gamma_t^R = -W_t^R$, the IC condition reduces to

$$\beta_t^R = \max\left\{0, \frac{b}{\theta} - W_t^R\right\}.$$

Thus, incentivizing the researcher is "free" when her continuation value is sufficiently high.[7] In this case, the total cost of incentivizing a two-agent team also depends on agents' continuation values, making the comparison between the two organizational structures more nuanced. However, several critical properties persist. Namely, the cost of incentivizing the two-agent team is independent of q. Meanwhile, the cost of incentivizing the solo innovator remains unchanged (recall that passive shirking was the binding constraint for the solo innovator). This implies that the *relative cost* of the solo innovator compared to a team still increases as lemons become harder to detect. As a result, when q is sufficiently small, the principal prefers team-based innovation. Whereas for q sufficiently large, a solo innovator is more attractive.

6.3 Development Only

Assumption 2 ensures that research is optimal. When it fails, the principal may find it optimal to save on the investment and incentive costs of the researcher and hire a single agent responsible for development and truthfully reporting.

Lemma 8 *Suppose the principal does not invest in research. Then the optimal contract that induces effort on development and truthful reporting is* $\beta = \gamma = b$.

The optimal contract with a solo developer is remarkably simple. Because the solo developer only controls the arrival of discoveries, there is no need to differentiate the incentives between lemons and peaches. In fact, under this regime, holding total compensation fixed, the principal prefers rewarding the agent with γ (promised utility) rather than β (transfers) for the same reason discussed in Section 6.1. Of course, truthfully reporting peaches requires that $\beta \geq \gamma$, which binds at the optimum.

As we show in the appendix, when θ is sufficiently high, a solo developer is dominated by a solo innovator for all continuation values and thus not part of the optimal arrangement. The converse is also true when θ is sufficiently small. A complete analysis of the optimal contract when the solo developer is neither dominant nor dominated is left for future work.

7 Conclusion

In this paper we study how a principal should organize innovation when progress requires two complementary tasks—research and development—and when agents both choose unob-

[7] This conclusion hinges on allowing the principal to punish one agent for a non-verifiable report made by another agent. If we impose the restriction that $\gamma_t^R \geq 0$, then this "simplified" version of the model collapses to our main specification.

servable effort and privately observe quality-relevant information during the process. Concentrating tasks in a solo innovator economizes on incentive payments when the agent has substantial continuation value, but it also creates a dynamic vulnerability to the double deviation in which the agent both reduces effort and misreports quality. Separating tasks across two agents eliminates this double deviation but introduces free riding, making research effort more expensive in a way that is constant over time. The resulting comparison yields a simple organizing principle: because the solo innovator's incentive cost rises as continuation value declines (especially when flawed output is hard to detect), the optimal organizational form can shift systematically over the project's life cycle.

Our main finding is that optimal organizational design can be *history dependent* even when the production technology is stationary. When flaws are easy to detect, the principal optimally relies on sequential generalists—a sequence of solo innovators responsible for both tasks. When flaws are difficult to detect, she instead hires a two-agent team immediately and sustains specialization throughout. For intermediate environments, the project begins with a solo innovator and then transitions to a two-agent structure via staggered overlap. The second hire may appear as a temporary takeover with the incumbent on standby, or as an external-consultant-style research specialist who joins an ongoing development effort; the consultant arrangement always arises as detection difficulty approaches the simultaneous-hiring threshold, while the standby arrangement requires that preserving the incumbent's stake be valuable at the margin. These regimes provide a unified explanation for common organizational patterns observed in innovation-intensive sectors.

Several avenues for future work are natural. Expanding the team beyond two agents or allowing endogenous monitoring and cross-checking technologies would let the principal trade off organizational design against investments that improve verifiability. Related, incorporating heterogeneity in agent abilities, learning about project difficulty over time, or explicit switching/reallocation costs would all further inform empirical predictions about who is hired, when expansions occur, and how task assignments evolve within the team. We view these directions as promising ways to build on the core insights developed here and to engage with additional institutional features of innovation-focused organizations.

Appendix

A Value Functions

This appendix derives the principal's value functions for a single agent and a two-agent team hired simultaneously. We begin with a useful observation: an ODE of the form

$$x_1 F'(W) + F(W) + W = x_0 \tag{A.1}$$

has the general solution

$$F(W) = C \exp\left(-\frac{W}{x_1}\right) - W + (x_0 + x_1) \tag{A.2}$$

where the constant C is determined by a boundary condition.

A.1 One Agent

Let $F_1(W)$ denote the principal's value function with one agent assigned to both tasks. Recall from Section 4 that $K \equiv \frac{b}{\theta q}$.

Region $W \leq \underline{W}$. In this region, the binding IC constraint is the double deviation, so $\beta_S(W) = K - W$. The continuation utility evolves as $dW_t = (K - W_t)(dN_t - dt)$. The HJB equation is:

$$0 = -2c - (K - W)F_1'(W) + [R - K - F_1(W)] \tag{A.3}$$

Rearranging: $(K - W)F_1'(W) + F_1(W) = \pi - K$, which has the linear solution

$$F_1(W) = AW + \pi - (1 + A)K$$

The boundary condition $F_1(0) = 0$ yields $A = \frac{\pi}{K} - 1 = \left(\frac{\theta q}{b}\right)\pi - 1$.

Region $W > \underline{W}$. Here $\beta_S = 2b$ is constant, so the HJB becomes:

$$\begin{aligned} 0 &= -2c - \beta_S F_1'(W) + [R - W - \beta_S - F_1(W)] \\ &= -2(c + b) - 2bF_1'(W) + [R - W - F_1(W)] \end{aligned} \tag{A.4}$$

Rearranging into form (A.1): $2bF_1'(W) + F_1(W) + W = \pi - 2b$. Using $x_1 = 2b$ and $x_0 = \pi - 2b$ (so that $x_0 + x_1 = \pi$) in (A.2):

$$F_1(W) = C_1 \exp\left(-\rho_1 W\right) - W + \pi, \qquad \rho_1 = \frac{1}{2b} \tag{A.5}$$

The boundary condition $F_1(\underline{W}) = A\underline{W} + \pi - (1+A)K$ (continuity at $\underline{W}$) determines:

$$C_1 = \exp(\rho_1 \underline{W})(1+A)(\underline{W} - K) \tag{A.6}$$

Optimal starting utility. The optimal initial utility satisfies $F_1'(W_1^*) = 0$, yielding:

$$W_1^* = \frac{\ln(-\rho_1 C_1)}{\rho_1}, \qquad F_1(W_1^*) = \pi - \frac{1}{\rho_1} - W_1^* \tag{A.7}$$

Remark 1 *If the contract uses randomization instead of rewarding γ for reporting a lemon (keeping $W \geq \widetilde{W}$ with $\widetilde{W} = \frac{2b(1-q)}{q}$), then $F_1'(0) = \left(\frac{q}{2b}\right)\pi - 1$. Rewarding for lemon reports dominates randomization if $\theta > 1/2$, which is Assumption 2. (The derivation follows steps analogous to those above and is omitted.)*

A.2 Two Agents Hired Simultaneously

With two agents hired simultaneously (one for each task), define $W = W_1 + W_2$ as total promised utility. The evolution is $dW_t = (\beta_D + \beta_R)(dN_t - dt)$ where $\beta_D + \beta_R = b(1 + 1/\theta)$. The HJB is:

$$0 = -2c - b\left(1 + \frac{1}{\theta}\right) F_2'(W) + \left[R - W - b\left(1 + \frac{1}{\theta}\right) - F_2(W)\right] \tag{A.8}$$

Rearranging into form (A.1) with $x_1 = \frac{b(1+\theta)}{\theta}$ and $x_0 = \pi - x_1$ (so that $x_0 + x_1 = \pi$). With boundary condition $F_2(0) = 0$, substituting into (A.2) gives $C = -(x_0 + x_1) = -\pi$:

$$F_2(W) = \pi(1 - \exp(-\rho_2 W)) - W, \qquad \rho_2 = \frac{\theta}{b(1+\theta)} < \rho_1 \tag{A.9}$$

Under Assumption 3, $F_2'(0) = \pi\rho_2 - 1 > 0$, so the optimal starting utility is interior:

$$W_{sim}^* = \frac{1}{\rho_2}\log(\pi\rho_2), \qquad F_2(W_{sim}^*) = \pi - \frac{1}{\rho_2} - W_{sim}^*$$

B Proofs

B.1 Proofs in Section 2

Proof of Lemma 1 Let $x \equiv \beta - \gamma$. Every feasible contract satisfies

$$x \geq \frac{b}{\theta}. \tag{A.10}$$

Because $\beta = \gamma + x$, the lower truthtelling constraint implies

$$\gamma \geq (1-q)(\gamma + x) \quad \Longleftrightarrow \quad \gamma \geq \frac{1-q}{q}x. \tag{A.11}$$

It follows that

$$U(\beta,\gamma) = \gamma + \lambda_1 x \geq \left(\frac{1-q}{q} + \lambda_1\right) x = \frac{D(q)}{q} x \geq \frac{bD(q)}{q\theta}. \tag{A.12}$$

Together with (Double-Shirking IC), this establishes the lower bound

$$U(\beta,\gamma) \geq \max\left\{2b, \frac{bD(q)}{q\theta}\right\}. \tag{A.13}$$

Now consider the contract in (6). It satisfies truthtelling by construction, and

$$\theta(\beta^* - \gamma^*) = q\theta\beta^* \geq b, \tag{A.14}$$

so the research-shirking constraint is satisfied. Its expected compensation is

$$U(\beta^*,\gamma^*) = D(q)\beta^* = \max\left\{2b, \frac{bD(q)}{q\theta}\right\}. \tag{A.15}$$

Thus, the contract attains the lower bound and is optimal.

Proof of Proposition 1 Starting with the assumptions that $\lambda_1 > 2\lambda_0$, so the deviation cutoff

$$\bar{q} = \frac{1}{1+\lambda_1-2\lambda_0} \tag{A.16}$$

lies strictly between zero and one. Let $D(q) \equiv \lambda_1 + (1-\lambda_1)(1-q)$. From the analysis of the solo operator, expected compensation is

$$C^{solo}(q) = D(q)\beta^* = \begin{cases} \dfrac{bD(q)}{q\theta}, & q < \bar{q}, \\ 2b, & q \geq \bar{q}. \end{cases} \tag{A.17}$$

For $q < \bar{q}$, the double deviation binds. For $q \geq \bar{q}$, the double-shirking constraint binds.

With two agents, the developer's expected compensation is b and the researcher's expected compensation is $\lambda_1 b/\theta$. Thus,

$$C^{team} = b + \frac{\lambda_1 b}{\theta} = \left(\frac{2\lambda_1 - \lambda_0}{\lambda_1 - \lambda_0}\right) b. \tag{A.18}$$

The team cost is independent of q because the researcher does not observe or report quality, while the developer's effort is already sunk when he observes it.

Define the organizational cutoff

$$q^* \equiv \frac{1}{1+\theta}. \tag{A.19}$$

In the region where the double deviation binds,

$$C^{solo}(q) - C^{team} = \frac{b}{\theta}\left(\frac{1}{q} - (1+\theta)\right). \tag{A.20}$$

Moreover, $q^* < \bar{q}$ so the organizational cutoff lies in the region where the double deviation binds. For $q \geq \bar{q}$,

$$C^{team} - C^{solo} = \frac{\lambda_0}{\theta} b > 0. \tag{A.21}$$

If $\lambda_1 < 2\lambda_0$, then $\bar{q} > 1$, so research shirking determines $C^{solo}(q)$ for every feasible $q \in (0, 1]$ and the same q^* applies.

B.2 Proof of Proposition 4

The proof of Proposition 4 works as follows. Through a series of lemmas, which partition the state space into regions, we construct the principal's value function under the conjectured optimal policy and then verify that it satisfies the HJB. In doing so, we illustrate the dynamics of how the continuation value of each agent evolves.

To simplify the mathematical expressions, we will solve for the optimal social welfare function $S(W_1, W_2) = V(W_1, W_2) + W_1 + W_2$, which for all $\vec{W} \in \mathcal{W}_2$ solves

$$S(\vec{W}) = \pi - \min\left\{\beta_S(W_2) S_2(\vec{W}), \beta_R S_2(\vec{W}) + \beta_D S_1(\vec{W})\right\}$$

subject to boundary condition $S(0, W_2) = F_1(W_2) + W_2, S(W_1, 0) = F_1(W_1) + W_1$, where $S_j \equiv \frac{\partial S(W_1, W_2)}{\partial W_j}$.

The verification arguments below all share a common structure, which we isolate in the following lemma.

Lemma A.1 (Verification) *Suppose that on some region the candidate social value function takes the multiplicative form*

$$S(\vec{W}) = \pi - \varphi_1(W_1)\,\varphi_2(W_2), \tag{A.22}$$

where φ_1, φ_2 are positive and decreasing, and let $\ell_i \equiv -\varphi_i'/\varphi_i \geq 0$ denote the associated log-decay rates, so that $S_i = \ell_i(W_i)\,\varphi_1(W_1)\varphi_2(W_2) > 0$. Then:

1. *Assigning both tasks to agent 2 weakly dominates the split assignment if and only if*

$$\beta_S(W_2) \;\leq\; \beta_R + \beta_D \frac{\ell_1(W_1)}{\ell_2(W_2)}, \tag{A.23}$$

   with the split assignment weakly dominant when the inequality is reversed.

2. *The principal is indifferent between assigning both tasks to agent 1 and assigning both tasks to agent 2 if and only if $\beta_S(W_1)\ell_1(W_1) = \beta_S(W_2)\ell_2(W_2)$.*

Proof. Both parts follow from $S_1/S_2 = \ell_1(W_1)/\ell_2(W_2)$: dividing $\beta_R S_2 + \beta_D S_1 \gtrless \beta_S(W_2)S_2$ through by $S_2 > 0$ gives part 1, and dividing $\beta_S(W_1)S_1 = \beta_S(W_2)S_2$ through by $\varphi_1\varphi_2 > 0$ gives part 2. ■

Every candidate value function constructed below takes the form (A.22), so each verification step reduces to computing ℓ_1 and ℓ_2 and checking (A.23).

Lemma A.2 *For $\vec{W}_t \in \mathcal{K}$, it is optimal to assign one agent to each task. The social value function is given by*

$$S(\vec{W}_t) = \left[1 - \exp\left(-\tau(\vec{W}_t)\right)\right]\pi \tag{A.24}$$

where $\tau(\vec{W}) \equiv \frac{W_1+W_2}{\beta_R+\beta_D}$. Any path such that one agent is assigned to each task and $\vec{W}_{t+s} \in \mathcal{K}$ for all $s \leq \tau(\vec{W})$ achieves this payoff.

Proof of Lemma A.2. The expression in (A.24) is derived from the conjecture that $\beta_R S_2 + \beta_D S_1 = \pi - S$ and the boundary condition $S(0,0) = 0$. Note that $\tau(\vec{W})$ corresponds to the length of time before $\vec{W}_t$ reaches the origin. To verify that assigning one task to each agent is optimal, note that $\tau(\vec{W}) = \rho_2(W_1 + W_2)$, so (A.24) has the form (A.22) with $\varphi_1 = \pi e^{-\rho_2 W_1}$ and $\varphi_2 = e^{-\rho_2 W_2}$, and hence $\ell_1 = \ell_2 = \rho_2$. By part 1 of Lemma A.1, the split assignment dominates if and only if $\beta_S(W_2) \geq \beta_R + \beta_D$, which holds because $W_2 \leq W_S$ for $\vec{W} \in \mathcal{K}$. Hence, the exact assignment is payoff irrelevant as long as the path does not exit $\mathcal{K}$. ■

Lemma A.3 *For any $\vec{W} \in \mathcal{A}_2$, it is optimal to assign both tasks to Agent 2 and*

1. *If $W_2 \leq \underline{W}$, the social value function is given by*

$$S(\vec{W}) = \pi - (K - W_2)c(W_1) \tag{A.25}$$

where

$$c(W_1) = \begin{cases} \left(\dfrac{1}{K - W_1/\theta}\right)\exp\left(-\dfrac{W_1}{\beta_D}\right)\pi & \text{if } W_1 \leq \theta W_S \quad \text{(A.26)} \\ \left(\dfrac{1}{\beta_D + \beta_R}\right)\exp\left(-\dfrac{W_1 + W_S}{\beta_D + \beta_R}\right)\pi & \text{if } W_1 \in (\theta W_S, W_S] \quad \text{(A.27)} \end{cases}$$

2. *If $W_2 > \underline{W}$, the social value function is given by*

$$S(\vec{W}) = \pi - k(W_1)e^{-\rho_1 W_2} \tag{A.28}$$

where $k(W_1) = \frac{c(W_1)}{\rho_1}e^{\rho_1 \underline{W}}$.

Proof. Conjecture that the solution involves assigning Agent 2 to both tasks in this region.

1. If $W_2 \leq \underline{W}$, $\beta_S(W_2) = K - W_2$ and the value function takes the form

$$S(\vec{W}) = \pi - \beta_S(W_2)S_2 \implies S = \pi - (K - W_2)c(W_1) \tag{A.29}$$

where $c(W_1)$ is a function of only W_1 that is determined from the boundary condition. Assigning Agent 2 to both tasks means moving vertically downward until hitting the boundary of $\mathcal{K}$. If $W_1 \leq \theta W_S$, this occurs when $W_2 = W_1/\theta$ at which point $S(W_1, W_1/\theta) = (1 - e^{-\frac{W_1}{\beta_D}})\pi$ (from (A.24)). Thus, the boundary condition is that

$$S\left(W_1, \frac{W_1}{\theta}\right) = \pi - \left(K - \frac{W_1}{\theta}\right) c(W_1) = \left(1 - e^{-\frac{W_1}{\beta_D}}\right) \pi \tag{A.30}$$

Solving for $c(W_1)$, we get the expression in (A.26). To verify that the conjectured policy is optimal, note that the candidate has the form (A.22) with $\varphi_1 = c(W_1)$ and $\varphi_2 = K - W_2$, so $\ell_2 = \frac{1}{K - W_2}$ and, from (A.26), $\ell_1 = \frac{1}{b} - \frac{1/\theta}{K - W_1/\theta}$. Condition (A.23) becomes

$$K - W_2 \;\leq\; \frac{b}{\theta} + (K - W_2)\left(1 - \frac{b/\theta}{K - W_1/\theta}\right) \quad\Longleftrightarrow\quad K - W_2 \;\leq\; K - \frac{W_1}{\theta}, \tag{A.31}$$

which holds for all $W_2 \geq \frac{W_1}{\theta}$, as required in $\mathcal{A}_2$. If $W_1 \in (\theta W_S, W_S]$ then the boundary condition applies at the top border of the kite region where $W_2 = W_S$, and

$$S(W_1, W_S) = \pi - (K - W_S)c(W_1) = \left[1 - \exp\left(-\frac{W_1 + W_S}{\beta_D + \beta_R}\right)\right] \pi. \tag{A.32}$$

Solving for $c(W_1)$, we get the expression in (A.27). Again the candidate has the form (A.22), now with $\ell_1 = \rho_2$ (from (A.27)), so condition (A.23) becomes

$$K - W_2 \;\leq\; \frac{b}{\theta} + b\rho_2(K - W_2) \quad\Longleftrightarrow\quad K - W_2 \;\leq\; \beta_D + \beta_R, \tag{A.33}$$

which holds because $W_2 \geq W_S$.

2. If $W_2 > \underline{W}$, $\beta_S(W_2) = 2b = \frac{1}{\rho_1}$. Therefore,

$$S = \pi - \beta_S(W_2)S_2 = \pi - \frac{S_2}{\rho_1} \;\Longrightarrow\; S = \pi - k(W_1)e^{-\rho_1 W_2} \tag{A.34}$$

Under the conjectured solution, the path moves vertically downward to the boundary $W_2 = \underline{W}$. Therefore, using (A.25), the boundary condition is $S(W_1, \underline{W}) = \pi - (K - \underline{W})c(W_1)$. Solving for $k(W_1)$ yields $k(W_1) = \frac{c(W_1)}{\rho_1} e^{\rho_1 \underline{W}}$. To verify the candidate satisfies the HJB, note that it has the form (A.22) with $\varphi_1 = c(W_1)$ and $\varphi_2 = \frac{e^{\rho_1 \underline{W}}}{\rho_1} e^{-\rho_1 W_2}$, so $\ell_2 = \rho_1$ and $\beta_S(W_2) = 2b = 1/\rho_1$. Condition (A.23) becomes

$$2b \;\leq\; \frac{b}{\theta} + 2b^2\,\ell_1(W_1).$$

For $W_1 < \theta W_S$, $\ell_1 = \frac{1}{b} - \frac{1/\theta}{K - W_1/\theta}$ and the condition reduces to $K - W_1/\theta \geq 2\beta_D$, which holds because $W_1 < \theta W_S$ implies $K - W_1/\theta > K - W_S = \beta_D + \beta_R > 2\beta_D$. For

$W_1 \in (\theta W_S, W_S]$, $\ell_1 = \rho_2$ and the condition reduces to $2 \leq \frac{1}{\theta} + \frac{2\theta}{1+\theta}$, which holds for all $\theta \leq 1$.

■

For $\vec{W} \in \mathcal{E}$, we break the proof up into three sub-regions of the space because the value function takes a different form in each sub-region.

Lemma A.4 *If both $W_1, W_2 \in [W_S, \underline{W}]$, it is optimal to assign both tasks to one agent but the exact assignment (both to Agent 1 or 2) is payoff irrelevant. The social value function is*

$$S(\vec{W}) = \pi - (K - W_2)(K - W_1)\rho_2 c(W_S) \tag{A.35}$$

where $c(W_S)$ is defined in (A.27).

Proof. Suppose both tasks are assigned to agent 1. The social value function is then

$$S(W_1, W_2) = \pi - \beta_S(W_1)S_1 \implies S = \pi - (K - W_1)\hat{c}(W_2) \tag{A.36}$$

for some function $\hat{c}$. The boundary condition is $W_1 = W_S$, where

$$S(W_S, W_2) = \pi - (K - W_S)\hat{c}(W_2) = \pi - (K - W_2)c(W_S) \tag{A.37}$$

where $c(W_S)$ is given by (A.27) with $W_1 = W_S$. Thus,

$$\hat{c}(W_2) = \left(\frac{K - W_2}{K - W_S}\right) c(W_S) = (K - W_2)\rho_2 c(W_S) \tag{A.38}$$

which then yields (A.35). The candidate (A.35) has the form (A.22) with $\ell_i = \frac{1}{K - W_i}$, so $\beta_S(W_i)\ell_i = 1$ for $i = 1, 2$ and part 2 of Lemma A.1 implies the principal is indifferent between assigning both tasks to agent 1 or agent 2; she achieves the same payoff assigning both tasks to agent 2 until reaching the boundary where $W_2 = W_S$. To verify the principal cannot do better by assigning one task to each agent, it suffices to consider the case where $W_1 \leq W_2$; multiplying condition (A.23) through by $K - W_1$ and dividing by $K - W_2$, it is equivalent to

$$K - W_1 \leq \beta_R \left(\frac{K - W_1}{K - W_2}\right) + \beta_D.$$

Observing that the left-hand side is at most $\beta_D + \beta_R$ (since $W_1 \geq W_S$), whereas the right-hand side is at least $\beta_R + \beta_D$ (since $W_1 \leq W_2$) completes the verification step. ■

Lemma A.5 *Suppose $W_2 > \underline{W}$ and $W_1 \in [W_S, \underline{W}]$, then assigning both tasks to either agent is optimal. The social value function is given by*

$$S(W_1, W_2) = \pi - \rho_2(K - W_1)\frac{c(W_S)}{\rho_1} e^{-\rho_1(W_2 - \underline{W})} \tag{A.39}$$

Proof. Suppose the principal assigns both tasks to Agent 2. So,

$$S = \pi - 2bS_2 \implies S = \pi - h(W_1)e^{-\rho_1 W_2} \tag{A.40}$$

for some $h(W_1)$ to be determined. The optimal path moves vertically downward to the boundary $W_2 = \underline{W}$. From (A.35), the boundary condition is $S(W_1, \underline{W}) = \pi - (K - W_1)(K - \underline{W})\rho_2 c(W_S)$. Solving yields $h(W_1) = (K - W_1)\,\rho_2\,c(W_S)\,\frac{e^{\rho_1 \underline{W}}}{\rho_1}$, which gives the expression stated in the lemma.

The candidate has the form (A.22) with $\varphi_1 \propto K - W_1$ and $\varphi_2 \propto e^{-\rho_1 W_2}$, so $\ell_1 = \frac{1}{K - W_1}$ and $\ell_2 = \rho_1$. Since $\beta_S(W_1)\ell_1 = 1 = 2b\rho_1 = \beta_S(W_2)\ell_2$, part 2 of Lemma A.1 implies the principal is indifferent between assigning both tasks to agent 1 or agent 2 (as conjectured above). Condition (A.23) reads

$$2b \;\leq\; \frac{b}{\theta} + \frac{2b^2}{K - W_1} \quad\Longleftrightarrow\quad 1 \;\leq\; \frac{1}{2\theta} + \frac{b}{K - W_1},$$

which holds because $W_1 \geq W_S$ implies $\frac{b}{K - W_1} \geq \frac{b}{K - W_S} = \frac{\theta}{1+\theta}$, and $\frac{1}{2\theta} + \frac{\theta}{1+\theta} \geq 1$ for all $\theta \leq 1$. This completes the verification. ■

Lemma A.6 *If $W_1, W_2 \geq \underline{W}$, assigning both tasks to either agent is optimal. The social value function is given by*

$$S(W_1, W_2) = \pi - Ce^{-\rho_1(W_1 + W_2)} \tag{A.41}$$

where

$$C = \frac{\rho_2 \exp(2\rho_1 \underline{W}) c(W_S)}{\rho_1^2} \tag{A.42}$$

Proof. Conjecture that both tasks are assigned to agent 2. Since $\beta_S(W_2) = 2b$, the conjecture implies that S takes the form $S(W_1, W_2) = \pi - C(W_1)e^{-\rho_1 W_2}$. By symmetry, $S(W_1, \underline{W}) = S(\underline{W}, W_1)$, and the latter is given by Lemma A.5 with the roles of the agents interchanged: $S(\underline{W}, W_1) = \pi - h(\underline{W})e^{-\rho_1 W_1}$, where $h(x) = (K - x)\,\rho_2\,c(W_S)\,\frac{e^{\rho_1 \underline{W}}}{\rho_1}$ is the coefficient function from the proof of Lemma A.5. The boundary condition at $W_2 = \underline{W}$ therefore implies $C(W_1) = h(\underline{W})e^{-\rho_1 W_1}e^{\rho_1 \underline{W}}$. Thus, $S = \pi - Ce^{-\rho_1(W_1 + W_2)}$, where

$$C = h(\underline{W})e^{\rho_1 \underline{W}} = \frac{\rho_2 \exp(2\rho_1 \underline{W}) c(W_S)}{\rho_1^2}. \tag{A.43}$$

Here $\ell_1 = \ell_2 = \rho_1$ and $\beta_S(W_1) = \beta_S(W_2) = 2b$, so part 2 of Lemma A.1 implies that whether both tasks are assigned to agent 1 or agent 2 is payoff irrelevant, and condition (A.23) holds because $2b \leq \beta_D + \beta_R$. ■

B.3 Characterization of G and W_2^*

Suppose that agent 1's continuation value is W_1 when agent 2 is hired. How much starting continuation value should that principal give to agent 2 and to what tasks should each agent be assigned? To answer these questions, recall that the principal's value function is denoted by $V(W_1, W_2)$. Let $W_2^*(W_1)$ denote the continuation value for agent 2 that solves

$$W_2^*(W_1) \in \arg\max_{W_2} V(W_1, W_2) \tag{A.44}$$

Denote $G(W_1) = V(W_1, W_2^*(W_1))$ as the principal's expected payoff conditional on hiring Agent 2 at $W_2 = W_2^*(W_1)$ when agent 1's utility is W_1.

To characterize W_2^* and G we will make use of several properties of V. The first property is that the principal's value function is linear in W_2 on $\underline{\mathcal{A}}_2 = \{\vec{W} \in \mathcal{A}_2 : W_2 < \underline{W}\}$ with slope $V_2 = c(W_1) - 1$, where $c(W_1)$ is decreasing in W_1. This property is analogous to the single agent value function below $\underline{W}$. Hence for any W_1 such that $c(W_1) > 1$, $W_2^*(W_1) > \underline{W}$. Let W_a be such that $c(W_a) = 1$. Since $c(0) = \pi/K$, we have $W_a > 0$ if and only if $\frac{\pi}{K} - 1 > 0$, which is equivalent to the project being worth initiating if there were only a single agent.

Lemma A.7 *If $K < \pi$, then there exists a $W_a > 0$ such that $(W_1, W_2^*(W_1)) \in \mathcal{A}_2$ for all $W_1 \in [0, W_a)$, while $(W_1, W_2^*(W_1)) \in \mathcal{K}$ for all $W_1 \in \mathcal{W}_1$ with $W_1 \geq W_a$. If $K \geq \pi$, then $(W_1, W_2^*(W_1)) \in \mathcal{K}$ for all $W_1 \in \mathcal{W}_1$.*

Proof. From (A.26) and (A.27), c is strictly decreasing with $c(0) = \pi/K$, so $W_a > 0$ exists if and only if $K < \pi$. For $W_1 < W_a$, $c(W_1) > 1$ and Lemma A.8 below shows $W_2^*(W_1) > \underline{W}$, so $(W_1, W_2^*(W_1)) \in \mathcal{A}_2$. For $W_1 \geq W_a$ (or for all W_1 when $K \geq \pi$), $c(W_1) \leq 1$, so $V_2 \leq 0$ everywhere above the kite region (see the first step of the proof of Lemma A.10), and the maximizer lies in $\mathcal{K}$. ■

Lemma A.8 *Suppose that $K < \pi$, then for all $W_1 \in [0, W_a)$*

$$W_2^*(W_1) = \frac{\ln(\rho_1 k(W_1))}{\rho_1} = \underline{W} + \frac{\ln(c(W_1))}{\rho_1} > \underline{W} \tag{A.45}$$

and

$$G(W_1) = \pi - W_1 - K - \frac{\ln(c(W_1))}{\rho_1} \tag{A.46}$$

Note that in this region, G is (1) decreasing in K (2) strictly concave in W for $W \in [0, \theta W_S]$, and (3) linear and decreasing for $W \in [\theta W_S, W_S]$. These will be useful properties for characterizing the optimal contract.

Proof. Because the value function is concave in W_2 and since $V_2 = c(W_1) - 1 > 0$ in the linear region, it is also strictly positive in the kite region. Hence, the only possible solution to (A.44) involves $W_2 > \underline{W}$. The first-order condition in the concave region (of W_2) yields the expression for W_2^*. Plugging the solution back into V yields the expression for G. ■

Next, we characterize the optimal W_2^* when $c(W_1) < 1$. Let $S_2^{kite}(W_1)$ denote the marginal social value of agent 2's continuation value as $\vec{W}$ approaches the upper boundary of the kite region from below (for $W_1 \leq W_S$):

$$S_2^{kite}(W_1) = \lim_{W_2 \uparrow \min\{W_1/\theta, W_S\}} S_2(W_1, W_2)$$

The following lemma shows that S_2 is discontinuous along the left boundary of the kite region but continuous along its top boundary.

Lemma A.9 *For $W_1 < \theta W_S$, $S_2^{kite}(W_1) > c(W_1)$. For $W_1 \in [\theta W_S, W_S]$, $S_2^{kite}(W_1) = c(W_1)$.*

Proof. Inside the kite region, $S_2(W_1, W_2) = \pi\rho_2 e^{-\rho_2(W_1+W_2)}$ by Lemma A.2, while $S_2 = c(W_1)$ in $\underline{\mathcal{A}}_2$. Along the left boundary, $W_2 = W_1/\theta$ and $\rho_2(W_1 + W_1/\theta) = W_1/b$, so $S_2^{kite}(W_1) = \pi\rho_2 e^{-W_1/b}$, whereas $c(W_1) = \pi e^{-W_1/b}/(K - W_1/\theta)$ by (A.26). Hence $S_2^{kite}(W_1) > c(W_1)$ if and only if $K - W_1/\theta > 1/\rho_2 = K - W_S$, i.e., $W_1 < \theta W_S$. Along the top boundary, $W_2 = W_S$ and $S_2^{kite}(W_1) = \pi\rho_2 e^{-\rho_2(W_1+W_S)} = c(W_1)$ by (A.27). ■

Lemma A.10 *Suppose $K < \pi$ and consider any $W_1 \geq W_a$*

1. *If $W_a < \theta W_S$, then*

$$W_2^*(W_1) = \begin{cases} W_1/\theta & W_1 \in (W_a, \theta W_{sim}^*/(1+\theta)) \\ W_{sim}^* - W_1 & W_1 \in (\theta W_{sim}^*/(1+\theta), W_{sim}^*/(1+\theta)) \end{cases} \tag{A.47}$$

and therefore

$$G(W_1) = \begin{cases} F_2(W_1(1+1/\theta)) & W_1 \in (W_a, \theta W_{sim}^*/(1+\theta)) \\ F_2(W_{sim}^*) & W_1 \in (\theta W_{sim}^*/(1+\theta), W_{sim}^*/(1+\theta)) \end{cases} \tag{A.48}$$

2. *If $W_a \in [\theta W_S, W_S)$, then for all $W_1 \in (W_a, W_{sim}^*/(1+\theta))$*

$$W_2^*(W_1) = W_{sim}^* - W_1 \tag{A.49}$$
$$G(W_1) = F_2(W_{sim}^*) \tag{A.50}$$

Proof. First, for W_1 such that $c(W_1) < 1$, the solution for $W_2^*(W_1)$ cannot lie above the kite region since the derivative is negative in both the linear strip and the concave strip. Hence, the solution must lie in $\mathcal{K}$ or below it.

For $W_1 < W_{sim}^*/(1+\theta)$, $V_2(W_1, \theta W_1) = F_2'(W_1(1+\theta)) = \pi\rho_2 e^{-\rho_2 W_1(1+\theta)} - 1 > \pi\rho_2 e^{-\rho_2 W_{sim}^*} - 1 = 0$ and hence $W_2^*(W_1) > \theta W_1$ implying the solution must lie in $\mathcal{K}$. Within $\mathcal{K}$, $V(W_1, W_2) = F_2(W_1 + W_2)$ which is increasing (decreasing) in W_2 for $W_1 + W_2 < (>) W_{sim}^*$. Hence, either (1) $W_2^*(W_1)$ lies on the upper boundary (i.e., $W_2^*(W_1) = \max\{W_2 : (W_1, W_2) \in \mathcal{K}\}$), which occurs when $W_1 + \max\{W_2 : (W_1, W_2) \in \mathcal{K}\} \leq W_{sim}^*$ or (2) $W_1 + W_2^*(W_1) = W_{sim}^*$. To complete the proof, it suffices to show that $W_1 + \max\{W_2 : (W_1, W_2) \in \mathcal{K}\} < W_{sim}^*$ if and only if $W_a < \theta W_S$ and $W_1 \in (W_a, \theta W_{sim}^*/(1+\theta))$.

Suppose that $W_a < \theta W_S$, then from (A.27), we have

$$c(\theta W_S) = \pi\rho_2 e^{-\rho_2(1+\theta)W_S} < 1 \implies \frac{1}{\rho_2}\ln(\rho_2\pi) = W_{sim}^* < (1+\theta)W_S \tag{A.51}$$

In other words, $W_a < \theta W_S$ is equivalent to the ray $W_2 = W_{sim}^* - W_1$ intersecting the kite region on its left boundary (i.e., where $\max\{W_2 : (W_1, W_2) \in \mathcal{K}\} = W_1/\theta$). Further, by Lemma A.9, $S_2^{kite}(W_a) > c(W_a) = 1$, so $V_2 > 0$ just inside the kite region at $(W_a, W_a/\theta)$, i.e., $F_2'(W_a(1+1/\theta)) > 0$, which implies that the point of intersection satisfies $\theta W_{sim}^*/(1+\theta) > W_a$. Hence, for all $W_1 \in (W_a, \theta W_{sim}^*/(1+\theta))$, $W_1 + \max\{W_2 : (W_1, W_2) \in \mathcal{K}\} = W_1 + W_1/\theta < W_{sim}^*$. Whereas, for $W_1 \in (\theta W_{sim}^*/(1+\theta), W_{sim}^*/(1+\theta)), \max\{W_2 : (W_1, W_2) \in \mathcal{K}\} \geq W_{sim}^*/(1+\theta)$.

Conversely, if $W_a \geq \theta W_S$ then $W^*_{sim} \geq (1+\theta)W_S$ implying the ray $W_2 = W^*_{sim} - W_1$ intersects the kite region along the top boundary (i.e., where $\max\{W_2 : (W_1, W_2) \in \mathcal{K}\} = W_S$). Because S_2 is continuous along the top boundary (Lemma A.9), $S_2(W_a, W_S) = c(W_a) = 1 \implies W_a + W_S = W^*_{sim}$ and hence for all $W_1 \in (W_a, W^*_{sim}/(1+\theta))$, $W_1 + \max\{W_2 : (W_1, W_2) \in \mathcal{K}\} = W_1 + W_S > W^*_{sim}$. ■

B.4 Proofs for Section 5.2

Proof of Lemma 5. If the active branch of V at $(w, W_2^*(w))$ is solo-1, then

$$V(w, W_2^*(w)) = \pi - (w + W_2^*(w)) - \beta_S(w)\big(1 + V_1(w, W_2^*(w))\big).$$

At a point contact with differentiable $\bar{G}$, we have $\bar{G}(w) = V(w, W_2^*(w))$ and, by the envelope theorem, $\bar{G}'(w) = V_1(w, W_2^*(w))$. Substituting into Φ yields

$$\Phi(w) = \pi - w - \beta_S(w)\big(1 + V_1\big) - \big[\pi - (w + W_2^*(w)) - \beta_S(w)(1 + V_1)\big] = W_2^*(w) \geq 0.$$

Thus w lies in the waiting region. Suppose instead that $\bar{G}$ is linear on an interval $[w_a, w_b]$. If the active branch at both $(w_a, W_2^*(w_a))$ and $(w_b, W_2^*(w_b))$ is solo-1, the argument above applied at w_a and w_b gives $\Phi(w_a) \geq 0$ and $\Phi(w_b) \geq 0$. Since Φ is constant on any interval where $\bar{G}$ is linear (see Case B in the proof of Lemma A.11), $\Phi(w) = \Phi(w_a) = \Phi(w_b) \geq 0$ for all $w \in [w_a, w_b]$. Hence every $w \in [w_a, w_b]$ is in the waiting region. ■

Lemma A.11 *Fix $K > 2b$ so that $\beta_S(W; K) = K - W$ on $W \in [0, W_S]$. Then, on $\mathcal{W}_1$:*

1. *$\Phi(W; K)$ is continuous and weakly increasing in W, and strictly increasing wherever $\bar{G}$ is strictly concave.*
2. *$\Phi(W; K)$ is continuous and strictly decreasing in K for every fixed W.*

Proof of Lemma A.11. Recall

$$\Phi(W; K) = \pi - W - (K - W)\big(1 + \bar{G}_W(W; K)\big) - \bar{G}(W; K),$$

with $\bar{G}$ the concave envelope of $G(w) = \max_{y \geq 0} V(w, y)$. Because $\bar{G}$ is concave and piecewise C^1, each point is either: (i) a *point contact* with G (so locally $\bar{G} = G$ and $\bar{G}_W = V_1(\cdot, W_2^*(\cdot; K))$), or (ii) lies on a *linear segment* $[w_a, w_b]$ with $\bar{G}(W; K) = c_0(K) + s(K)W$ and constant slope $s \in \partial\bar{G}$ given by the chord between two supporting endpoints where $\bar{G} = G$.

Weak monotonicity in W. Differentiate w.r.t. W wherever $\bar{G}$ is C^2 (point contact), and use the onesided derivative on linear pieces:

$$\Phi_W(W; K) = -(K - W)\,\bar{G}_{WW}(W; K).$$

Since $\bar{G}$ is concave, $\bar{G}_{WW} \leq 0$, hence $\Phi_W \geq 0$, with *strict* inequality wherever $\bar{G}_{WW} < 0$. On any linear segment, $\bar{G}_{WW} \equiv 0$, so $\Phi_W \equiv 0$ there. Continuity in W follows because $\bar{G}$ and $\bar{G}_W$ match continuously at junctions of concave arcs and linear pieces.

Strict monotonicity in K. We show $\Phi_K(W;K) < 0$ for every $W \in \mathcal{W}_1$.
Case A: point contact at (W,K). Here $\bar{G} = G$ locally and

$$\Phi_K(W;K) = -\big(1+\bar{G}_W\big) \;-\; (K-W)\,\bar{G}_{WK} \;-\; \bar{G}_K. \tag{$*$}$$

By the envelope, $\bar{G}_W = V_1(W, W_2^*(W;K))$, and the twoagent HJB yields the slope bound $1+V_1 > 0$ at every finite state, so the first term in $(*)$ is *strictly negative*. If the supporting region is $\mathcal{K}$ ("kite"), then $\bar{G}$ is locally independent of K, so $\bar{G}_K = \bar{G}_{WK} = 0$ and hence $\Phi_K = -(1+\bar{G}_W) < 0$. If the supporting region is $\mathcal{A}_2$, direct differentiation of the closed form gives

$$\Phi_K(W;K) = -\Big(1 + \frac{2b\big(W + K\theta^2 - 2\theta W\big)}{(W-K\theta)^2}\Big) \;<\; 0.$$

Thus $\Phi_K < 0$ at all point-contact states in $\mathcal{K} \cup \mathcal{A}_2$.
Case B: linear segment at W. Let $[w_a(K), w_b(K)]$ be the supporting chord with $\bar{G}(W;K) = c_0(K) + s(K)W$ for $W \in [w_a, w_b]$. Using $\beta_S(W;K) = K - W$, we have

$$\Phi(W;K) = \pi - W - (K-W)\big(1+s(K)\big) - \big(c_0(K) + s(K)W\big) = \pi - K - c_0(K) - s(K)K,$$

so $\Phi(\cdot;K)$ is *constant in* W on $[w_a, w_b]$. In particular, for any interior $W \in (w_a, w_b)$,

$$\Phi_K(W;K) = \Phi_K\big(w_a(K);K\big),$$

because $\Phi_W \equiv 0$ on the segment and the endpoints move continuously in K. At the endpoint $w_a(K)$ we have *point contact* ($\bar{G} = G$), and by construction of $\mathcal{W}_1$ the supporting endpoints of a segment used at $W \in \mathcal{W}_1$ lie in $\mathcal{K} \cup \mathcal{A}_2$. Therefore $\Phi_K\big(w_a(K);K\big) < 0$ by Case A, implying $\Phi_K(W;K) < 0$ throughout the linear segment.

Combining Cases A and B yields $\Phi_K(W;K) < 0$ for every $W \in \mathcal{W}_1$. ■

Proof of Lemma 6. We first claim that it is sufficient to show that (i) there exists a K_1 such that $\Phi(0;K_1) > 0$ and (ii) there exists K_2 such that $\Phi(\max(\mathcal{W}_1);K_2) < 0$. We then prove (i) and (ii).

Suppose that (i) and (ii) hold.

- For (1), $\Phi(0;K_1) > 0$ (by assumption) and $\Phi(0;K_2) < 0$ because we found K_2 such that $\Phi(\max(\mathcal{W}_1);K_2) < 0$ and $\Phi_W \geq 0 \Rightarrow \Phi(\max(\mathcal{W}_1)) \geq \Phi(0)$. Since Φ is continuous in K, the intermediate value theorem implies that there exists K_{seq} such that $\Phi(0;K_{\text{seq}}) = 0$. Because Φ is decreasing in K, $\Phi(0;K) > 0$ for all $K < K_{\text{seq}}$.

- For (3), by monotonicity in W, $\Phi(\max(\mathcal{W}_1);K_1) \geq \Phi(0;K_1) > 0 > \Phi(\max(\mathcal{W}_1);K_2)$. The intermediate value theorem guarantees the existence of K_{sim}. Monotonicity in K and W ensure that Φ is negative on $\mathcal{W}_1$ for all $K > K_{\text{sim}}$.

- For (2), for all $K \in (K_{\text{seq}}, K_{\text{sim}})$, $\Phi(0;K) < 0$ and $\Phi(\max(\mathcal{W}_1);K) > 0$. Continuity and monotonicity of Φ with respect to W_1 ensure that $\hat{W}_1$ is interior for all such K.

To prove (i), notice that if $K \leq \beta_R+\beta_D$, then $W_S \leq 0$ and $\mathcal{K} = \mathcal{A}_i = \emptyset$, so $(0, W_2^*(0)) \in \mathcal{E}$. By Lemma 5, $\Phi(0) = W_2^*(0) > 0$, where the strict inequality follows from Assumption 3, which ensures the project is worth initiating with a single agent since $K \leq \beta_R + \beta_D < \pi$.

To prove (ii), take any $K_2 > \pi$. By Lemma A.7, $(W_1, W_2^*(W_1)) \in \mathcal{K}$ for all $W_1 \in \mathcal{W}_1$, so G is given by the kite-region expressions in Lemma A.10 (with $W_a = 0$), and $\bar{G} = G$ is constant and equal to $F_2^{\max} \equiv F_2(W_{sim}^*)$ on the flat portion ending at $\max(\mathcal{W}_1)$. On that portion $\bar{G}' = 0$, so $\Phi(\max(\mathcal{W}_1); K_2) = \pi - K_2 - F_2^{\max} < 0$, where the inequality holds because $K_2 > \pi > \pi - F_2^{\max}$. Since Φ is weakly increasing in W, $\Phi(\,\cdot\,; K_2) < 0$ on all of $\mathcal{W}_1$.

In fact, Lemma A.12 below establishes that $\bar{G}$ always attains its maximum $F_2^{\max}$ on this flat portion at (and above) the threshold, so the identity $K_{\text{sim}} = \pi - F_2^{\max}$ holds without further qualification. Finally, we verify that $K_{\text{seq}} < K_{\text{sim}}$ holds strictly. If $\bar{G}(\,\cdot\,; K_{\text{seq}})$ coincides with G on part of the $\mathcal{A}_2$ branch or of the kite branch, then it is strictly concave there (see the properties following Lemma A.8), so $\Phi(\max(\mathcal{W}_1); K_{\text{seq}}) > \Phi(0; K_{\text{seq}}) = 0$ and hence $K_{\text{sim}} > K_{\text{seq}}$. The only alternative is that $\bar{G}(\,\cdot\,; K_{\text{seq}})$ consists of a single linear segment from $W = 0$ to the flat portion of G; because the envelope is tangent to G at any contact point where G is differentiable, and the flat portion has slope zero, such a segment would have to be horizontal at height $F_2^{\max}$, which requires the knife-edge coincidence $G(0; K_{\text{seq}}) = F_2^{\max}$. Outside this degenerate case, $K_{\text{seq}} < K_{\text{sim}}$. ■

Lemma A.12 *Suppose $K > \beta_R + \beta_D$ (so that $\mathcal{K} \neq \emptyset$) and let $K^\circ \equiv \pi - F_2^{\max}$, where $F_2^{\max} \equiv F_2(W_{sim}^*) > 0$. Then:*

1. $\max_{W \in \mathcal{W}_1} \bar{G}(W; K) < \pi - K$ *for all $K < K^\circ$;*
2. $\max_{W \in \mathcal{W}_1} \bar{G}(W; K) = F_2^{\max}$ *for all $K \geq K^\circ$, attained on the flat portion of G identified in Lemma A.10;*
3. $K_{\text{sim}} = K^\circ$.

Proof. Since the concave envelope preserves the maximum, $\max_{\mathcal{W}_1} \bar{G} = \max_{\mathcal{W}_1} G$. Every value of G is of one of two types. If $(W, W_2^*(W)) \in \mathcal{K}$, then $G(W)$ is a kite-region value $F_2(\cdot) \leq F_2^{\max} = \pi - K^\circ$. If $(W, W_2^*(W)) \in \mathcal{A}_2$—that is, $W < W_a$, so $c(W) > 1$—then (A.46) directly gives

$$\pi - K - G(W) \;=\; W + \frac{\ln c(W)}{\rho_1} \;>\; 0, \tag{A.52}$$

where the inequality is strict even at $W = 0$ because $c(0) = \pi/K > 1$ whenever $W_a > 0$. Thus the $\mathcal{A}_2$-branch values of G never reach $\pi - K$, *even when they exceed* $F_2^{\max}$.

For part 1, take $K < K^\circ$: kite values are at most $\pi - K^\circ < \pi - K$ and $\mathcal{A}_2$-branch values are below $\pi - K$ by (A.52), so $\max G < \pi - K$.

For part 2, take $K \geq K^\circ$ and note first that the flat portion is nonempty: $W_S(K) = K - (\beta_R+\beta_D) \geq K^\circ - (\beta_R+\beta_D) = W_{sim}^* > W_{sim}^*/(1+\theta)$, so $W_{sim}^* < (1+\theta)W_S$, which is precisely the configuration of case 1 in Lemma A.10 (here we use $K^\circ = \frac{1}{\rho_2} + W_{sim}^*$ and $\beta_R + \beta_D = \frac{1}{\rho_2}$). On the flat portion $G = F_2^{\max}$, while $\mathcal{A}_2$-branch values are below $\pi - K \leq \pi - K^\circ = F_2^{\max}$ by (A.52); hence $\max G = F_2^{\max}$.

For part 3, by part 2 the envelope satisfies $\bar{G} = F_2^{\max}$ on the flat portion for $K \geq K^\circ$ (it majorizes $G = F_2^{\max}$ there and its maximum equals $F_2^{\max}$), so $\bar{G}' = 0$ there and $\Phi(\max(\mathcal{W}_1); K) = \pi - K - F_2^{\max}$, which is zero at $K = K^\circ$ and negative for $K > K^\circ$. For $K \in (\beta_R + \beta_D, K^\circ)$, suppose toward a contradiction that $\Phi(\max(\mathcal{W}_1); K) \leq 0$; since Φ is weakly increasing in W, immediate hiring would then be weakly optimal from every state, so the principal's value would equal $\max \bar{G}$. But part 1 and the construction in Case 2 of the proof of Lemma 7 below (which uses only part 1) produce a strategy with payoff strictly above $\max \bar{G}$—a contradiction. Hence $\Phi(\max(\mathcal{W}_1); K) > 0$ for all $K < K^\circ$ and $K_{\text{sim}} = K^\circ$. ■

Proof of Lemma 7. Write $M(K) \equiv \max_{W \in \mathcal{W}_1} \bar{G}(W; K)$ and recall $K_{\text{sim}} = \pi - F_2^{\max}$ (Lemma A.12).

Case 1: $K \leq \beta_R + \beta_D$. Here $\Phi(0; K) > 0$ (see the proof of Lemma 6), and since Φ is weakly increasing in W, the stopping region is empty and $\hat{W}_1 = 0$. By (33), using $\bar{G}(0) = G(0) = F_1(W_1^*) = \pi - K - \frac{\ln(\pi/K)}{\rho_1}$ from (A.7),

$$A(0) = \frac{\pi - K - \bar{G}(0)}{K} = \frac{\ln(\pi/K)}{\rho_1 K} > 0,$$

since $K \leq \beta_R + \beta_D < \pi$ by Assumption 3. Hence $W_0^* > 0 = \hat{W}_1$.

Case 2: $\beta_R + \beta_D < K < K_{\text{sim}}$. By Lemma A.12 (part 1), $\pi - K > M(K)$. Choose a maximizer p of $\bar{G}$ and note $p \in \mathcal{W}_1 \subseteq [0, W_S]$ with $W_S < \underline{W}$. For any initial promise W_0 and hiring date τ, since $\beta_S(W) \geq K - W$, the flow payoff in (30) satisfies $\pi - W_t - \beta_S(W_t) \leq \pi - K$, with equality if and only if $W_t \leq \underline{W}$. Hiring immediately therefore yields at most $M(K)$ from any state, whereas starting at $W_0 = p + \delta$ for small $\delta > 0$ and hiring agent 2 when W_t reaches p, at date $\tau_\delta > 0$, yields

$$\Pi(\tau_\delta,\, p + \delta) = (\pi - K)\big(1 - e^{-\tau_\delta}\big) + e^{-\tau_\delta} M(K) \;>\; M(K),$$

because the path remains below $\underline{W}$ so the flow equals $\pi - K$ exactly. Hence $\tau^* > 0$: agent 1 is hired strictly before agent 2.

Case 3: $K \geq K_{\text{sim}}$. By Lemma A.12 (part 2), $M(K) = F_2^{\max}$, and $\pi - K \leq \pi - K_{\text{sim}} = F_2^{\max}$. The same flow bound gives, for every (τ, W_0),

$$\Pi(\tau, W_0) \;\leq\; (\pi - K)\big(1 - e^{-\tau}\big) + e^{-\tau} M(K) \;\leq\; M(K),$$

with the first inequality strict whenever $\tau > 0$ and $K > K_{\text{sim}}$. Stopping immediately at a maximizer of $\bar{G}$—which by Lemma A.12 (part 2) lies on the flat portion, i.e., in the kite region with total promised utility W_{sim}^*—attains $M(K)$. Hence $W_0^* = \hat{W}_1$ and the two agents are hired simultaneously. ■

Proof of Proposition 6. On $(K_{\text{seq}}, K_{\text{sim}})$, Lemma 6 implies that the hiring threshold $\hat{W}_1(K) \in (0, \max(\mathcal{W}_1))$ is interior. We first establish that $\hat{W}_1(K)$ is weakly increasing in K, without appealing to differentiability. Fix $K < K'$. By Lemma A.11, $\Phi(\hat{W}_1(K); K') < \Phi(\hat{W}_1(K); K) \leq 0$, so $\hat{W}_1(K)$ lies in the stopping region at K', and hence $\hat{W}_1(K') \geq \hat{W}_1(K)$. (Where $\bar{G}$ is strictly concave at the contact point, $\Phi_W > 0$ and the implicit function theorem

applied to $\Phi(\hat{W}_1(K); K) = 0$ yields the strict version $\hat{W}_1'(K) = -\Phi_K/\Phi_W > 0$; where $\bar{G}$ is linear, Φ is constant in W and $\hat{W}_1(K)$ may jump upward.)

Since $K_{\text{sim}} \leq \pi$, Lemma A.7 guarantees that for every $K \in (K_{\text{seq}}, K_{\text{sim}})$ there exists a boundary point $W_a(K) \in (0, \max(\mathcal{W}_1))$ such that

$$(W_1, W_2^*(W_1)) \in \mathcal{A}_2 \text{ for } W_1 < W_a(K), \qquad (W_1, W_2^*(W_1)) \in \mathcal{K} \text{ for } W_1 > W_a(K).$$

Hence, for any $K \in (K_{\text{seq}}, K_{\text{sim}})$:

$$\text{Standby Developer} \iff \hat{W}_1(K) < W_a(K), \qquad \text{External Consultant} \iff \hat{W}_1(K) > W_a(K).$$

Because $c(\,\cdot\,; K)$ is decreasing in K, the boundary point $W_a(K)$ is decreasing in K, while $\hat{W}_1(K)$ is weakly increasing; hence $\hat{W}_1(K) - W_a(K)$ is increasing in K and the sets

$$\{K \in (K_{\text{seq}}, K_{\text{sim}}) : \hat{W}_1(K) < W_a(K)\}, \qquad \{K \in (K_{\text{seq}}, K_{\text{sim}}) : \hat{W}_1(K) > W_a(K)\}$$

are intervals. By the definition of $\tilde{K}$, the former is $(K_{\text{seq}}, \tilde{K})$, on which $(\hat{W}_1(K), W_2^*(\hat{W}_1(K))) \in \mathcal{A}_2$ and the Standby Developer structure is optimal, while the latter contains $(\tilde{K}, K_{\text{sim}})$, on which $(\hat{W}_1(K), W_2^*(\hat{W}_1(K))) \in \mathcal{K}$ and the External Consultant structure is optimal. This establishes the interval structure underlying both cases of the proposition.

Next, we show that $\tilde{K} < K_{\text{sim}}$. At $K = K_{\text{sim}}$, Lemma A.12 implies that $\bar{G} = F_2^{\max}$ on the flat portion of G, while every value of G to its left is strictly below $F_2^{\max}$: the $\mathcal{A}_2$-branch values by (A.52) (using $\pi - K_{\text{sim}} = F_2^{\max}$), and the kite-branch values because $F_2(x) < F_2^{\max}$ for $x < W_{sim}^*$. Now observe that at any contact point where G is differentiable, the concave envelope is tangent to G (since $\bar{G} - G$ is minimized there). The kite branch is differentiable with slope zero at the left endpoint of the flat portion, so a linear segment of $\bar{G}$ ending there would have to be horizontal at height $F_2^{\max}$—impossible, as its left contact point would require $G = F_2^{\max}$ strictly to the left of the flat portion. Hence $\bar{G}$ coincides with the strictly concave kite branch on an interval adjacent to the flat portion, on which $\Phi(\,\cdot\,; K_{\text{sim}})$ is strictly increasing up to $\Phi = 0$ on the flat portion. Consequently $\Phi(W; K_{\text{sim}}) < 0$ for all W below that interval; in particular $\Phi(W_a; K_{\text{sim}}) < 0$, since $W_a < \theta W_{sim}^*/(1+\theta)$ by Lemma A.10 (whose case 1 applies at $K = K_{\text{sim}}$, as shown in the proof of Lemma A.12). By continuity of Φ in K, $\Phi(W_a(K); K) < 0$ for all K sufficiently close to K_{sim}, so $\hat{W}_1(K) > W_a(K)$ and the External Consultant structure obtains near K_{sim}. Hence $\tilde{K} < K_{\text{sim}}$.

Both cases now follow: if $K_{\text{seq}} < \tilde{K} < K_{\text{sim}}$, the two intervals in the statement are $(K_{\text{seq}}, \tilde{K})$ and $(\tilde{K}, K_{\text{sim}})$; if $\tilde{K} = K_{\text{seq}}$, the External Consultant interval is all of $(K_{\text{seq}}, K_{\text{sim}})$. ■

We now provide a necessary and sufficient condition for the Standby Developer interval to be nonempty. To make the dependence on the sequential threshold explicit, we write $G(\,\cdot\,; K_{\text{seq}})$, $W_a(K_{\text{seq}})$, and $W_S(K_{\text{seq}})$ for the corresponding objects evaluated at $K = K_{\text{seq}}$.

Condition 1 *Let* $x_a \equiv \left(1 + \frac{1}{\theta}\right) W_a(K_{\text{seq}})$ *and define* $\Lambda(x) \equiv G(0; K_{\text{seq}}) + G'(0; K_{\text{seq}}) \frac{\theta}{1+\theta} x - F_2(x)$.

1. $G'(0; K_{\text{seq}}) > 0$.
2. $\Lambda(x) > 0$ *for all* $x \in [x_a, W_{sim}^*]$.

Lemma A.13 *Suppose that $W_a(K_{\text{seq}}) < \theta\, W_S(K_{\text{seq}})$. Then Condition 1 holds if and only if $\tilde{K} > K_{\text{seq}}$.*

Proof. Throughout, evaluate all objects at $K = K_{\text{seq}}$ and, to ease notation, suppress this argument; let $\ell(W) \equiv G(0) + G'(0)W$ denote the tangent line to G at $W_1 = 0$. Under the hypothesis, case 1 of Lemma A.10 applies, so G consists of the strictly concave $\mathcal{A}_2$ branch (A.46) on $[0, W_a)$; the kite portion $G(W_1) = F_2(x(W_1))$, where $x(W_1) \equiv (1 + \frac{1}{\theta})W_1$, on $[W_a, \theta W^*_{sim}/(1+\theta)]$; and the flat portion $G = F_2^{\max}$ on $[\theta W^*_{sim}/(1+\theta), \max(\mathcal{W}_1)]$. Note that $\ell(W) - G(W) = \Lambda(x(W))$ on the kite portion.

Step 1: Condition 1 holds if and only if $\bar{G} = G$ in a neighborhood of 0. Suppose Condition 1 holds. Then ℓ majorizes G strictly on $(0, \max(\mathcal{W}_1)]$: on the $\mathcal{A}_2$ branch by strict concavity and tangency at 0; on the kite portion because $\ell - G = \Lambda > 0$ by part 2; and on the flat portion because part 1 makes ℓ increasing, so $\ell(W) \geq \ell(\theta W^*_{sim}/(1+\theta)) = F_2^{\max} + \Lambda(W^*_{sim}) > F_2^{\max}$. If $\bar{G}$ did not coincide with G near 0, it would contain a linear segment from 0 to a contact point $w_b > 0$ with slope $s = (G(w_b) - G(0))/w_b$; since $\bar{G} \geq G$ near 0 and G is tangent to ℓ at 0, we must have $s \geq G'(0)$, whence $G(w_b) \geq \ell(w_b)$—contradicting strict majorization.

Conversely, suppose $\bar{G} = G$ on a neighborhood of 0, so that $\bar{G}(0) = G(0)$ and $\bar{G}'(0) = G'(0)$. Substituting the closed forms $G'(0) = 1 - \frac{2b}{\theta K_{\text{seq}}}$ and $G(0) = \pi - K_{\text{seq}} - \frac{\ln(\pi/K_{\text{seq}})}{\rho_1}$ (both from (A.46)) into the free-boundary condition $\Phi(0; K_{\text{seq}}) = 0$ that defines K_{seq} yields

$$K_{\text{seq}} \;=\; \frac{2b}{\theta} + \frac{\ln(\pi/K_{\text{seq}})}{\rho_1} \;>\; \frac{2b}{\theta}, \tag{A.53}$$

where the inequality uses $K_{\text{seq}} < K_{\text{sim}} < \pi$; hence part 1 holds. For part 2, suppose toward a contradiction that $\Lambda(\hat{x}) \leq 0$ for some $\hat{x} \in [x_a, W^*_{sim}]$, and let $\hat{w} = \theta\hat{x}/(1+\theta)$. Then $G(\hat{w}) \geq \ell(\hat{w})$, so the chord from $(0, G(0))$ to $(\hat{w}, G(\hat{w}))$ has slope at least $G'(0)$ and therefore lies strictly above the $\mathcal{A}_2$ branch near 0 (strict concavity again); since $\bar{G}$ majorizes any chord between two points of G, this gives $\bar{G} > G$ near 0—a contradiction. Hence part 2 holds as well.

Step 2: $\bar{G} = G$ near 0 implies $\tilde{K} > K_{\text{seq}}$. On a neighborhood of 0, $\Phi_W = -(K - W)\bar{G}'' > 0$ by strict concavity of the $\mathcal{A}_2$ branch, so there exists $w_0 \in (0, W_a(K_{\text{seq}}))$ with $\Phi(w_0; K_{\text{seq}}) > \Phi(0; K_{\text{seq}}) = 0$. Since $\Phi(w_0; \cdot)$ is continuous in K (Lemma A.11) and $W_a(K)$ is continuous in K (it is the root of the jointly continuous, strictly decreasing $c(\cdot\,; K) = 1$), for all $K > K_{\text{seq}}$ sufficiently close to K_{seq} we have $\Phi(w_0; K) > 0$ and $w_0 < W_a(K)$. Hence $\hat{W}_1(K) < w_0 < W_a(K)$, the Standby Developer structure obtains, and $\tilde{K} > K_{\text{seq}}$.

Step 3: $\bar{G} \neq G$ near 0 implies $\tilde{K} = K_{\text{seq}}$. In this case $\bar{G}$ contains a linear segment from 0 to a contact point $w_b > 0$. The segment cannot end on the $\mathcal{A}_2$ branch: tangency at $w_b \in (0, W_a)$ would require $G'(w_b) = s \geq G'(0)$, contradicting the strictly decreasing derivative of the branch (and a chord ending at a branch point without tangency would lie below G near 0, so it could not belong to a majorant of G). Hence $w_b \geq W_a(K_{\text{seq}})$. Because $\Phi(\cdot\,; K_{\text{seq}})$ is constant on linear segments of $\bar{G}$ (Lemma A.11), $\Phi(W; K_{\text{seq}}) = \Phi(0; K_{\text{seq}}) = 0$ for all $W \in [0, w_b]$, and by strict monotonicity of Φ in K, $\Phi(W; K) < 0$ on $[0, w_b]$ for every $K > K_{\text{seq}}$. Therefore $\hat{W}_1(K) \geq w_b \geq W_a(K_{\text{seq}}) \geq W_a(K)$ for all $K \in (K_{\text{seq}}, K_{\text{sim}})$, the Standby Developer structure never obtains, and $\tilde{K} = K_{\text{seq}}$. ∎

Remark 2 (Verifying Condition 1) *Condition 1 is verifiable in closed form. By (A.46), $G'(0;K_{\text{seq}}) = 1 - \frac{2b}{\theta K_{\text{seq}}}$ and $G(0;K_{\text{seq}}) = \pi - K_{\text{seq}} - \frac{\ln(\pi/K_{\text{seq}})}{\rho_1}$, where, by (A.53), K_{seq} solves the scalar equation $\rho_1 K_{\text{seq}} + \ln K_{\text{seq}} = \frac{1}{\theta} + \ln \pi$ whenever the condition holds. Part 1 is then equivalent to $K_{\text{seq}} > 2b/\theta$, i.e., to the detection probability that generates $K = K_{\text{seq}}$ lying below $1/2$. Since F_2 is concave, Λ is convex with stationary point $x^\circ = \frac{1}{\rho_2} \ln \frac{\pi\rho_2}{1+G'(0;K_{\text{seq}})\theta/(1+\theta)}$, so part 2 reduces to the single inequality $\Lambda\big(\min\{\max\{x^\circ, x_a\}, W^*_{sim}\}\big) > 0$.*

B.5 Proofs for Section 6

Proof of Proposition 7. Fix $\epsilon \in (0, \theta)$. We prove optimality within each organizational regime using the principal's HJB and the incentive constraints.

Solo innovator. Let W^S denote the promised continuation utility to the solo innovator. To implement full effort and truthful reporting, it is sufficient that (β, γ) satisfy:

$$
\begin{aligned}
&(1-\epsilon)\beta + \epsilon\gamma \geq 2b && \text{(double shirking)},\\
&(\theta-\epsilon)(\beta-\gamma) \geq b && \text{(passive shirking)},\\
&\theta\beta \geq b && \text{(active shirking)},\\
&\gamma \geq (1-q)\beta - qW^S && \text{(truthful lemons)},\\
&\beta \geq \gamma && \text{(truthful peaches)}.
\end{aligned}
$$

Note that double shirking and truthful peaches jointly imply $\beta \geq 2b$, so active shirking is implied whenever $\theta \geq 1/2$ (Assumption 2) and we may drop it in what follows. Moreover, once truthful lemons holds, any deviation that combines shirking with misreporting is weakly dominated by the corresponding shirk-and-report-truthfully deviation, so the constraints above are sufficient for full effort and truthful reporting. Define $K_\epsilon := \frac{b}{q(\theta-\epsilon)}$ and $(\bar\beta_\epsilon, \bar\gamma_\epsilon)$ as in the statement, and set

$$
\widehat{W}_\epsilon \;:=\; K_\epsilon - \bar\beta_\epsilon \;=\; \frac{(1-\epsilon q)b}{q(\theta-\epsilon)} - 2b.
$$

(The threshold $\widehat{W}_\epsilon$, at which the truthful-lemons constraint starts to bind, is distinct from the cost-comparison threshold $\bar W_\epsilon$ defined in Corollary 1.) To justify which constraints bind, consider the principal's HJB under on-path effort, written in terms of the solo-innovator value F:

$$
F(W^S) = \max_{(\beta,\gamma)\in\Gamma(W^S)} \Big\{ -2c - \big((1-\epsilon)\beta + \epsilon\gamma\big)\, F'(W^S) + (1-\epsilon)\big(R - W^S - \beta\big) + \epsilon\, F(W^S + \gamma) \Big\},
$$

where $\Gamma(W^S)$ is the set defined by the IC constraints above. Assuming F is concave and $1 + F'(W) \geq 0$ (total surplus increases with W), the objective is weakly decreasing in β holding γ fixed (since $\partial/\partial\beta = -(1-\epsilon)(1+F'(W^S)) \leq 0$), and weakly decreasing in γ holding β fixed (since $\partial/\partial\gamma = \epsilon(F'(W^S+\gamma) - F'(W^S)) \leq 0$ by concavity). Thus the principal wants to keep total incentives as small as possible. When the double-shirking constraint binds, the principal can trade off β for γ while keeping $(1-\epsilon)\beta + \epsilon\gamma = 2b$; along that locus, the

principal prefers shifting weight toward γ (since marginal cost of increasing β is -1 while the marginal cost of increasing γ is $F'(W^S+\gamma) \geq -1$).

Low W^S ($W^S < \widehat{W}_\epsilon$). Suppose truthful lemons were slack at the optimum. Then the logic above pins down the contract from the double-shirking and passive-shirking constraints alone, yielding the constant pair $(\bar{\beta}_\epsilon, \bar{\gamma}_\epsilon)$ derived in the high-W^S case below; but that pair violates truthful lemons precisely when $W^S < \widehat{W}_\epsilon$—a contradiction. Hence truthful lemons binds. With $\gamma = (1-q)\beta - qW^S$, passive-shirking becomes $(\theta-\epsilon)\big(\beta - (1-q)\beta + qW^S\big) = q(\theta-\epsilon)(\beta+W^S) \geq b$, so the smallest feasible β is $\beta = K_\epsilon - W^S$ and $\gamma = (1-q)K_\epsilon - W^S$.

High W^S ($W^S \geq \widehat{W}_\epsilon$). When W^S is high, truthful lemons can be slack. Then the HJB logic above implies that the optimal contract minimizes incentives subject to double shirking and passive shirking, with the tradeoff between β and γ governed by these constraints. In particular, the optimal pair has $(1-\epsilon)\beta + \epsilon\gamma = 2b$ and $\beta - \gamma = b/(\theta-\epsilon)$ (double shirking and passive shirking bind), which solves to the constant pair $(\bar{\beta}_\epsilon, \bar{\gamma}_\epsilon)$. Truthful lemons is slack precisely when $W^S \geq \widehat{W}_\epsilon$, i.e., $\bar{\gamma}_\epsilon \geq (1-q)\bar{\beta}_\epsilon - qW^S$. Combining the two regions gives the max-formulas stated in the proposition.

Two-agent team. Let (W_D, W_R) be promised utilities. The researcher's passive-shirking IC is $(\theta-\epsilon)(\beta^R_\epsilon - \gamma^R_\epsilon) \geq b$. From the two-agent HJB, concavity implies the principal weakly prefers smaller continuation-value rewards for the researcher, so the optimal choice sets $\gamma^R_\epsilon = 0$ and makes the constraint bind: $\beta^R_\epsilon = b/(\theta-\epsilon)$.

For the developer, effort requires $(1-\epsilon)\beta^D_\epsilon + \epsilon\gamma^D_\epsilon \geq b$ and truthful peaches requires $\beta^D_\epsilon \geq \gamma^D_\epsilon$. Along a binding effort constraint, the HJB implies the principal prefers shifting weight from transfers toward continuation utility, but $\beta^D_\epsilon \geq \gamma^D_\epsilon$ limits this substitution, so the optimum sets $\beta^D_\epsilon = \gamma^D_\epsilon = b$. Truthful lemons is then slack for all $W_D \geq 0$ because $\gamma^D_\epsilon = b \geq (1-q)b - qW_D$. Thus the stated two-agent contract implements effort and truthful reporting and is optimal within the two-agent regime. ■

B.6 Sufficient Conditions for Research

In this section, we derive a sufficient condition under which the principal always finds it optimal to implement effort on research. We begin with the proof of Lemma 8, which characterizes the optimal contract that implements effort on development and truthful reporting.

Proof of Lemma 8. Ignoring terms independent of (β, γ), the principal solves

$$\max_{\beta,\gamma} -(1-\theta)\beta F'(W) - \theta\gamma F'(W) - (1-\theta)\beta + \theta F(W+\gamma)$$

subject to

$$(1-\theta)\beta + \theta\gamma \geq b, \qquad \beta \geq \gamma, \qquad \gamma \geq (1-q)\beta - qW.$$

Let $\delta = (1-\theta)\beta + \theta\gamma$ and rewrite the objective as $-\delta F'(W) - (\delta - \theta\gamma) + \theta F(W+\gamma)$. Since $F'(W) \geq -1$, the objective is decreasing in δ (its derivative in δ, holding γ fixed, is $-(1+F'(W)) \leq 0$), so the constraint $\delta \geq b$ binds. Holding $\delta = b$ fixed, the objective is increasing in γ (its derivative is $\theta\big(1+F'(W+\gamma)\big) \geq 0$), so γ increases until $\beta \geq \gamma$ binds, giving $\beta = \gamma = \delta = b$. The remaining constraint $\gamma \geq (1-q)\beta - qW$ is then satisfied. ■

Allowing for a solo developer, the principal's HJB in any state becomes:

$$V(w_1, w_2) = \max\{S_1,\, S_2,\, RD,\, DR,\, D_1,\, D_2\}(w_i, w_j),$$

where S_i denotes the *solo-innovator* branch for agent i, D_i denotes the *solo-developer* branch for agent i, and RD (DR) denotes the branch in which agent 1 is the researcher (developer) and agent 2 is the developer (researcher). Our goal is to show that for sufficiently high θ, the principal never finds it optimal to use a solo-developer branch. It suffices to prove that S_i dominates D_i for all (w_i, w_j). A similar comparison using the two-agent team (RD or DR) yields the condition $\theta \geq 1 - c/R$ (precisely Assumption 2), which is more demanding than the solo-innovator bound derived below. Hence the team comparison does not tighten the sufficient condition.

Lemma A.14 *Suppose V is concave in each argument. Then the solo-innovator branch (S_i) dominates the solo-developer branch (D_i) at every state if*

$$(1-\theta)\left[R - \min\left\{\frac{1}{2}, \theta q\right\}(R - 2c)\right] \leq c.$$

Proof. Fix w_j and write $v(w_i) = V(w_i, w_j)$. Let

$$\beta_0 \equiv \beta_S(0) = \max\left\{2b, \frac{b}{\theta q}\right\}.$$

From S_i,

$$v(w_i) = (R - 2c) - (w_i + w_j) - \beta_S(w_i)\big(1 + v'(w_i)\big).$$

Evaluating at $w_i = 0$ and using $v(0) = F_1(w_j) \geq 0$,

$$\beta_0\big(1 + v'_+(0, w_j)\big) = (R - 2c) - (w_j + F_1(w_j)) \;\leq\; R - 2c,$$

so

$$1 + V_i(w_i, w_j) \;=\; 1 + v'(w_i) \;\leq\; 1 + v'_+(0, w_j) \;\leq\; \frac{R - 2c}{\beta_0}.$$

In the solo-developer branch (Lemma 8),

$$D_i = (1-\theta)\big(R - (w_i + w_j + b)\big) - c - b\, V_i(w_i, w_j) + \theta V(w_i + b, w_j).$$

By concavity in w_i,

$$V(w_i + b, w_j) \leq V(w_i, w_j) + b\, V_i(w_i, w_j),$$

so

$$D_i \leq V(w_i, w_j) + (1-\theta)\,\tilde{D}_i - c, \qquad \tilde{D}_i := (R - (w_i + w_j + b)) - V - bV_i.$$

Using the identity for the solo operator, $V + \beta_S(w_i)V_i = R - 2c - (w_i + w_j) - \beta_S(w_i)$, we obtain

$$\begin{aligned}\tilde{D}_i &= 2c + (\beta_S(w_i) - b)\big(1 + V_i(w_i, w_j)\big) \le 2c + (\beta_0 - b)\frac{R - 2c}{\beta_0} \\ &= R - \frac{b}{\beta_0}(R - 2c) = R - \min\left\{\frac{1}{2}, \theta q\right\}(R - 2c) =: D_{\max}(\theta, q),\end{aligned}$$

where the second equality follows from

$$\frac{b}{\beta_0} = \min\left\{\frac{1}{2}, \theta q\right\}.$$

Thus, for every (w_i, w_j),

$$D_i(w_i, w_j) \le V(w_i, w_j) + (1 - \theta)D_{\max}(\theta, q) - c.$$

It follows that $S_i \ge D_i$ everywhere whenever

$$(1 - \theta)\left[R - \min\left\{\frac{1}{2}, \theta q\right\}(R - 2c)\right] \le c,$$

which is the claimed condition. ■

References

Admati, A. R. and M. Perry (1991). Joint projects without commitment. *Review of Economic Studies 58*(2), 259–276.

Bolton, P. and C. Harris (1999). Strategic experimentation. *Econometrica 67*(2), 349–374.

Bonatti, A. and J. Horner (2011). Collaborating. *American Economic Review 101*(2), 632–663.

Brown, T. (2008). Design thinking. *Harvard Business Review 86*(9), 62–72.

Cetemen, D., I. Hwang, and A. Kaya (2020). Uncertainty-driven cooperation. *Theoretical Economics 15*(3), 1023–1058.

Che, Y.-K., E. Iossa, and P. Rey (2021). Prizes versus contracts as incentives for innovation. *Review of Economic Studies 88*(5), 2149–2178.

Che, Y.-K. and S.-W. Yoo (2001). Optimal incentives for teams. *American Economic Review 91*(3), 525–541.

Georgiadis, G. (2015). Projects and team dynamics. *American Economic Review 81*(1), 187218.

Gershkov, A. and M. Perry (2012). Dynamic contracts with moral hazard and adverse selection. *Review of Economic Studies 79*(1), 268–306.

Green, B. and C. R. Taylor (2016). Breakthroughs, deadlines, and self-reported progress: Contracting for multistage projects. *American Economic Review 106*(12), 3660–99.

Gromb, D. and D. Martimort (2007). Collusion and the organization of delegated expertise. *Journal of Economic Theory 137*(1), 271–299.

Halac, M., N. Kartik, and Q. Liu (2016). Optimal contracts for experimentation. *Review of Economic Studies 83*(3), 1040–1091.

He, Z., B. Wei, J. Yu, and F. Gao (2017). Optimal long-term contracting with learning. *Review of Financial Studies 30*(6), 2006–2065.

Iossa, E. and D. Martimort (2012). Risk allocation and the costs and benefits of public–private partnerships. *RAND Journal of Economics 43*(3), 442–474.

Iossa, E. and D. Martimort (2015). The simple microeconomics of public-private partnerships. *Journal of Public Economic Theory 17*(1), 4–48.

Keller, G., S. Rady, and M. Cripps (2005). Strategic experimentation with exponential bandits. *Econometrica 73*(1), 39–68.

Laux, C. (2001). Limited-liability and incentive contracting with multiple projects. *RAND Journal of Economics*, 514–526.

Levy, S. (2011). *In the Plex: How Google Thinks, Works, and Shapes Our Lives*. New York: Simon & Schuster.

Liu, B. and J. Lu (2018). Pairing provision price and default remedy: optimal two-stage procurement with private r&d efficiency. *RAND Journal of Economics 49*(3), 619–655.

Martimort, D. and J. Pouyet (2008). To build or not to build: Normative and positive theories of public–private partnerships. *International Journal of Industrial Organization 26*(2), 393–411.

Marx, L. M. and S. A. Matthews (2000). Dynamic voluntary contribution to a public project. *Review of Economic Studies 67*(2), 327–358.

O'Reilly, C. A. and M. L. Tushman (2004). The ambidextrous organization. *Harvard Business Review 82*(4), 74–81.

Pavlak, A. (2004). Project troubleshooting: Tiger teams for reactive risk management. *Project Management Journal*.

Schmitz, P. W. (2005). Allocating control in agency problems with limited liability and sequential hidden actions. *RAND Journal of Economics*, 318–336.

Varas, F. (2018). Managerial short-termism, turnover policy, and the dynamics of incentives. *Review of Financial Studies 31*(9), 3409–3451.

Yildirim, H. (2006). Getting the ball rolling: Voluntary contributions to a large-scale public project. *Journal of Public Economic Theory 8*(4), 503–528.

Yildirim, H. (2023). Who fares better in teamwork? *RAND Journal of Economics 54*(2), 299–324.